# Vortex rings and rows on the hyperbolic plane: one stability boundary through the horocycle

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**Abstract**

We study the linear stability of equally spaced point vortices of equal strength on the hyperbolic plane $\mathbb{H}^2$, with the true Green function $-\log\tanh(d/2)$ and no background vorticity (Kimura's model). For the regular $N$-ring the Hessian of the augmented Hamiltonian splits into $2\times 2$ Fourier blocks whose entries are rational in $\xi=\tanh^2(r/2)$, $r$ the hyperbolic radius. From them we prove: rings with $N\le 7$ are stable at every radius; every $N\ge 8$ has exactly one threshold radius, below which the ring is unstable in the mode $\lfloor N/2\rfloor$ and above which it is spectrally stable; and the threshold is an algebraic number, certified to 45 digits for $8\le N\le 16$. For odd $N\ge 9$ there is a short window (width $\sim 10^{-8}$–$10^{-7}$ in $\xi$ for $N\le 15$, $O(N^{-3})$ in general) where the ring is spectrally stable but the Hessian is indefinite: the critical mode carries two eigenvalues of opposite Krein signature, so a definiteness (energy–momentum) criterion misses stability there. For large $N$, $\cosh r_*(N)=N/\tau_\infty+O(1/N)$ and the arc spacing at threshold tends to $\ell_*=2\pi/\tau_\infty=0.843908\ldots$, which is exactly the critical spacing of an infinite row on a horocycle: that row is stable iff $\ell\ge\ell_*$, with no Krein window. Rows along hypercycles of geodesic curvature $\kappa$ are unstable at every spacing iff $|\kappa|\le 1/\sqrt{2}$; above it they are stable beyond one threshold $\ell_{\mathrm{hyp}}(\kappa)$ (an interval-arithmetic certificate shows the threshold is unique), real-analytic on $(1/\sqrt{2},1)$, which blows up at $1/\sqrt{2}$ and ends at the horocycle point $(1,\ell_*)$, where the ring thresholds also accumulate. The boundary does not continue analytically through that point: the hypercycle branch has a square-root branch point there and the even and odd ring thresholds arrive with different finite slopes. Most statements are checked in Lean 4 with Mathlib; the rest are labelled as certified (interval or exact arithmetic), derived or numerical.

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# 1 Introduction

**The problem.** A point vortex on a surface moves in the velocity field of the others; the motion is Hamiltonian, with the Green function of the Laplacian as pair interaction and the area form as symplectic form. In the plane, the regular ring of $N$ equal vortices (Thomson's polygon) is linearly stable for $N \le 6$, neutral for $N = 7$ and unstable for $N \ge 8$ [3], and the infinite straight row (a single von Kármán row) is unstable at every spacing [17, 8, 1]. On a curved surface the ring has an extra parameter, its radius measured against the curvature, and the question becomes: *for which radii is the ring stable?* On the hyperbolic plane the question extends naturally to rows: equally spaced vortices along any curve of constant geodesic curvature $\kappa$ (a geodesic, a hypercycle, a horocycle or a circle) form a relative equilibrium, translating along the curve.

**The model.** We use the Green function of the Laplace–Beltrami operator of $\mathbb{H}^2$ (curvature $-1$) that vanishes at infinity,

$$G(d) = -\frac{1}{2\pi} \log \tanh \frac{d}{2},$$

$d$ the geodesic distance. Since $\mathbb{H}^2$ is non-compact and homogeneous no compensating background vorticity is needed, the Robin function is constant, and the Hamiltonian is the plain pair sum $H = -\sum_{j<k} \log \tanh(d_{jk}/2)$ (unit circulations, factor $2\pi$ absorbed). This is the model attributed to Kimura [7]; it is the case $\lambda = -1$ of the image Green function of Montaldi–Tokieda [11, §4.2], and Boatto [2] uses the same Green function. It differs from the "sinh model" $-\log \sinh(d/2)$ studied by Montaldi and Tokieda [11], which carries a uniform background vorticity (§8). The tanh interaction is screened beyond one curvature radius, $-\log \tanh(d/2) \approx 2e^{-d}$, and this changes the large-$N$ behaviour completely.

**Results.** Write $\xi = \tanh^2(r/2) \in (0,1)$ for a ring of hyperbolic radius $r$ (in the Poincaré disk $\xi$ is the squared Euclidean radius), and $\ell$ for the arc length between neighbours along the carrying curve.

**A.** *Rings* (§§3–4). The Hessian of $H - \Omega J$ at the ring splits into explicit $2\times 2$ blocks $M_m$, rational in $\xi$ (Theorem 3.1). Rings with $N \le 7$ are spectrally stable, with positive-definite Hessian on the non-rotational modes, at every radius; every $N \ge 8$ has exactly one threshold $\xi_*(N)$, with critical mode $\lfloor N/2 \rfloor$ (Theorem 3.6); Table 1 lists $\xi_*(N)$, $8 \le N \le 16$. For odd $N \ge 9$ the definiteness threshold $\xi_{\mathrm{EM}}(N)$ lies strictly above $\xi_*(N)$; in between, the critical mode has two eigenvalues of opposite Krein signature (Theorem 4.2).

**B.** *Large $N$ and the horocycle row* (§5). $\cosh r_*(N) = N/\tau_\infty + O(1/N)$, where $\tau_\infty = 7.44533924\ldots$ is the zero of an explicit function, and the threshold spacing $\ell_N \to \ell_* = 2\pi/\tau_\infty$ (Theorem 5.2). The infinite row on a horocycle is stable iff $\ell \ge \ell_*$ and its Bloch symbol is the limit of the ring blocks (Theorem 5.4).

**C.** *Hypercycle rows* (§6). A row on a hypercycle of curvature $\kappa$ is unstable at every spacing iff $|\kappa| \le \kappa_c = 1/\sqrt{2}$ (Theorem 6.4); for $|\kappa| > \kappa_c$ it is stable exactly for $\ell \ge \ell_{\mathrm{hyp}}(\kappa)$, with $\ell_{\mathrm{hyp}}$ real-analytic, tending to $+\infty$ at $\kappa_c$ and to $\ell_*$ at $\kappa = 1$ (Theorem 6.6; uniqueness of the threshold is certified by interval arithmetic). The threshold in strip spacing, $\ell_{\mathrm{hyp}}\sqrt{1-\kappa^2}$, is strictly decreasing in $\kappa$; that $\ell_{\mathrm{hyp}}$ itself decreases is seen numerically. The critical mode is always the pairing mode $q = \pi$. Appendix D proves the modular (dual) form of the threshold function that the certificate and §7 use.

**D.** *The horocycle point* (§7). The ring thresholds accumulate only at $(1, \ell_*)$; the hypercycle branch reaches it with a square-root branch point in $\kappa$ and a flat remainder, the even and odd rings with two different finite slopes, so no analytic curve continues the boundary through the horocycle (Theorem 7.1).

Figure 1 shows the resulting stability boundary in the $(\kappa, \ell)$ plane.

**Why curvature stabilises.** The planar row is unstable at every spacing, and so is a row along a geodesic of $\mathbb{H}^2$, although it has the same screened interaction (Theorem 6.4(a)). What stabilises sparse horocycle and hypercycle rows is the bending of the carrying curve: the critical curvature $1/\sqrt{2}$ is where, at large spacing, the stiffness that the curvature gives the pairing mode balances the attraction of a vortex to its displaced neighbours (Remark 6.5). For rings the same mechanism appears as a critical *spacing*: a large ring is stable once neighbours are more than $\ell_* \approx 0.844$ curvature radii apart.

**Status labels.** Every claim carries one label. **[Lean]**: checked in Lean 4 with Mathlib [16]; we name the theorem in the library `HyperbolicRings` (directory `lean/`). **[Proved]**: proof by hand in this paper. **[Certified]**: exact rational or interval (Arb [6]) arithmetic, with script and output named. **[Derived]**: formal asymptotics. **[Numerical]**: floating-point evidence. Where a Lean statement rests on a modelling input that is not itself formalised (Hamilton's equations in a chart, the identification of the linearisation of an infinite row with a lattice sum of pair terms) we say so. The file `README.md` next to the paper maps every numbered statement to its proof, Lean theorem, script and output.

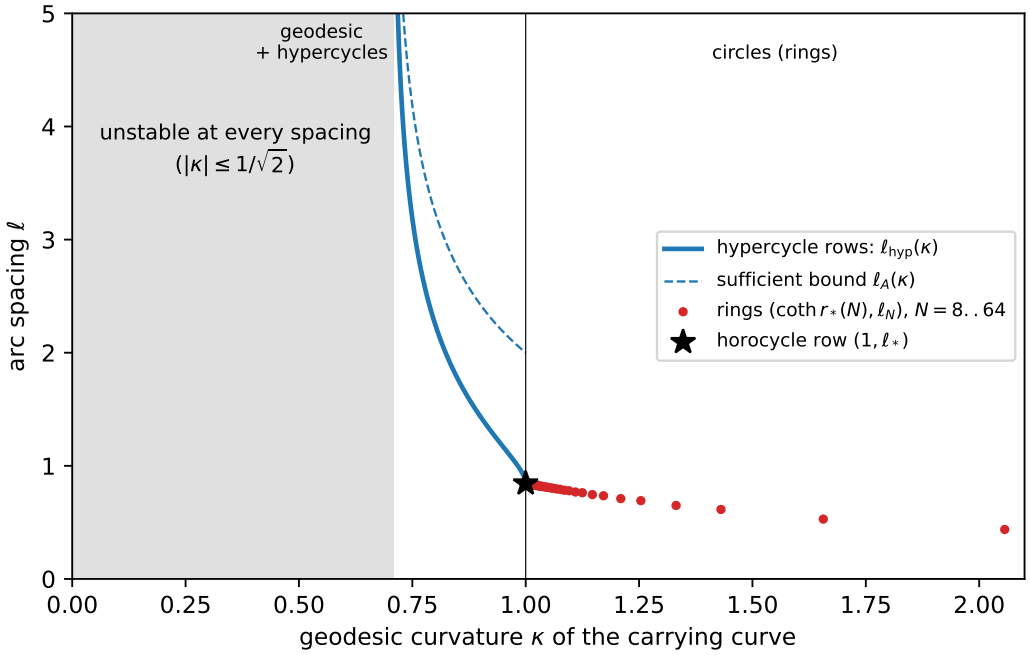



Figure 1: Stability boundary of equally spaced vortex arrays on $\mathbb{H}^2$ (tanh model) in the plane of geodesic curvature $\kappa$ of the carrying curve and arc spacing $\ell$. Grey: unstable at every spacing (Theorem 6.4). Blue: hypercycle threshold $\ell_{\mathrm{hyp}}(\kappa)$ (Theorem 6.6); dashed: the explicit sufficient bound $\ell_A(\kappa)$. Star: the horocycle row (Theorem 5.4). Red: threshold rings $(\coth r_*(N), \ell_N)$, $N = 8, \dots, 64$. Arrays above the curve are spectrally stable. Plotted values are floating point (`code/make_figures.py`).

## 2 Setting

### 2.1 Hamiltonian and stability notions

In the Poincaré disk $\mathbb{D}$ with metric $4|dz|^2/(1-|z|^2)^2$ we have $\tanh(d(z,w)/2) = |z-w|/|1-z\bar{w}|$, so

$$H = \sum_{j<k} \Big( -\log|z_j - z_k| + \log|1 - z_j\bar{z}_k| \Big), \qquad \omega = \sum_k \frac{4\,dx_k \wedge dy_k}{(1-|z_k|^2)^2}, \qquad J = \sum_k \cosh r_k, \tag{1}$$

where $J$ generates the rotations about 0. (The function $f(d) = \log\tanh(d/2)$ satisfies $f'' + \coth d\, f' = 0$, so $G$ is the Green function; the pair term in (1) is also the Green function of the flat unit disk with a conducting wall, but the symplectic form differs.) The full symmetry group is the isometry group $PSL(2,\mathbb{R})$.

A configuration $x_0$ is a *relative equilibrium* if it is a critical point of $F = H - \Omega J$ for some $\Omega$ (here $J$ is the momentum of a one-parameter subgroup: rotations for rings, the translations along the curve for rows). The linearised flow at $x_0$ is $\mathcal{L} = \omega^{-1} d^2F(x_0)$.

**Definition 2.1.** $x_0$ is *spectrally stable* if every eigenvalue of $\mathcal{L}$ has zero real part. It is *definite* (on a given invariant subspace $V$) if $d^2F(x_0)|_V$ is positive definite. For a ring, $V$ is the sum of the Fourier modes $m = 1, \dots, N-1$ (all but the rotation/dilation mode $m = 0$). For an infinite row, stability and definiteness are imposed on each Bloch wave $q \in (0, 2\pi)$.

Definiteness on $V$ is the hypothesis of the energy–momentum method [13]; it implies spectral stability on $V$ but not conversely. When $d^2F|_V$ is indefinite and the spectrum is imaginary, the eigenvalues carry *Krein signatures* (the sign of $d^2F$ on the eigenvector) [9]: an eigenvalue pair of opposite signatures can collide and leave the imaginary axis under perturbation, and nonlinear stability is not decided by the quadratic form. We claim nonlinear stability nowhere in this paper. (Positive definiteness on $V$ yields Lyapunov stability only modulo the symmetry group; the modes $m = \pm 1$ are tangent to the non-compact boost directions of $PSL(2,\mathbb{R})$, and we have not verified the hypotheses of a specific energy–momentum theorem for this model.)

### 2.2 The ring

Let $w = e^{2\pi i/N}$ and $z_j = \sqrt{\xi}\, w^j e^{s_j + i\theta_j}$, so the ring of Euclidean radius $\sqrt{\xi}$ is $s = \theta = 0$. Then $\omega = \sum_j \rho(s_j)\, ds_j \wedge d\theta_j$ with $\rho_0 := \rho(0) = 4\xi/(1-\xi)^2 = \sinh^2 r$. Put

$$c := \cosh r = \frac{1+\xi}{1-\xi}, \qquad X := \xi^N, \qquad W_k := \sum_{\substack{p \geq 1 \\ p \equiv k \ (N)}} p\, \xi^p = \frac{k\xi^k}{1-X} + \frac{N\xi^{k+N}}{(1-X)^2} \quad (1 \leq k \leq N).$$

**Lemma 2.2** (relative equilibrium; **[Lean]**). *For every $N \geq 1$ and $\xi \in (0,1)$ the ring is a critical point of $F = H - \Omega J$ in all $2N$ variables exactly when $\Omega\rho_0 = -\frac{N-1}{2} + \frac{\xi}{1-\xi} - \frac{NX}{1-X}$. Lean:* `HyperbolicRings.Tanh.ring_critical` *(with* `ring_critical_iff`*).*

The proof uses the power-sum expansion $\log|1-u| = -\operatorname{Re}\sum_{p\geq 1} u^p/p$, which turns (1) into

$$H = -\sum_{j<k} \log|z_j - z_k| - \frac{1}{2}\sum_j \log(1-|z_j|^2) - \frac{1}{2}\sum_{p\geq 1} \frac{|P_p|^2}{p}, \qquad P_p = \sum_j z_j^p; \tag{2}$$

on a ring the last term is diagonal in the Fourier modes. The $\theta$-derivatives vanish by symmetry; the $s_j$-derivatives give the stated $\Omega$. (Appendix A gives the computation; Lean derives it from the Hamiltonian.)

## 3 Fourier blocks and the ring threshold

**Theorem 3.1** (Fourier blocks; **[Lean]**). *At the ring, $d^2F$ is block-circulant in $(s,\theta)$. On the Fourier mode $(s_j,\theta_j) = (S,T)w^{mj}$, $1 \leq m \leq N-1$, it acts as the Hermitian matrix $M_m = \left(\begin{smallmatrix} a_m & ig_m \\ -ig_m & b_m \end{smallmatrix}\right)$, where*

$$a_m = (N-1)c - \tfrac{1}{2}(c-1)^2 - \tfrac{m(N-m)}{2} + \frac{2NcX}{1-X} - NW_N - \tfrac{N}{2}(W_m + W_{N-m}),$$
$$b_m = \tfrac{m(N-m)}{2} + NW_N - \tfrac{N}{2}(W_m + W_{N-m}), \qquad g_m = \tfrac{N}{2}(W_m - W_{N-m}).$$

*All entries are rational functions of $\xi$ with rational coefficients; $a_{N-m} = a_m$, $b_{N-m} = b_m$, $g_{N-m} = -g_m$, and $g_{N/2} = 0$ for even $N$. Lean:* `HyperbolicRings.Tanh.blocks_eigen` *(the Hessian, computed from $H$ as explicit circulant matrices in* `BlocksHess`*, acts on $(w^{mb})_b$ by $a_m, b_m, g_m$),* `blocks_hermitian`*.*

Here $a_m$ is the radial and $b_m$ the tangential stiffness of mode $m$, and $g_m$ is a radial–tangential coupling which has no analogue in the planar or sinh-model ring. The hand proof is in Appendix A. Under $\xi \mapsto 1/\xi$ the entries satisfy $a_m \mapsto a_m$, $b_m \mapsto b_m$, $g_m \mapsto -g_m$ (inversion in the boundary circle), so the cleared numerators are self-reciprocal polynomials; we use this only as a check.

**Proposition 3.2** (linearisation; **[Lean]**). *On mode $m$ the linearised flow is $\rho_0(\dot S,\dot T)^\top = J_0 M_m (S,T)^\top$, $J_0 = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, with eigenvalues $\rho_0\lambda = -ig_m \pm \sqrt{-a_m b_m}$; the spectrum of $\mathcal{L}$ on $\mathbb{R}^{2N}$ is the union over $m$. Hence the ring is spectrally stable iff $a_m b_m \geq 0$ for all $1 \leq m \leq N-1$, and definite on $V$ iff $a_m > 0$ and $a_m b_m - g_m^2 > 0$ for all such $m$. Lean:* `linFlow_spectral_iff`, `hessQ_posDef_compl_iff` *(module* `Tanh.LinearFlow`*),* `flowBlock_hasEigenvalue_iff`, `specStable_iff_spectral`.

*Proof.* $\det(J_0 M_m - \lambda) = (\lambda + ig_m)^2 + a_m b_m$; a Hermitian $2\times 2$ matrix is positive definite iff its $(1,1)$ entry and its determinant are positive. Distinct modes are $d^2F$- and $\omega$-orthogonal. □

So $g_m$ shifts the frequencies but does not affect spectral stability; it does affect definiteness (§4).

### 3.1 The exact symbol

Put $\tau := -\frac{N}{2}\log\xi \in (0,\infty)$, so that $\xi^N = e^{-2\tau}$ and $\cosh r = \coth(\tau/N)$ ($\tau \to 0$ as $r \to \infty$), and for $1 \leq m \leq N-1$ put $x := 1 - 2m/N \in (-1,1)$. Let

$$Q(x,\tau) = \cosh\tau\cosh x\tau - x\sinh\tau\sinh x\tau, \qquad R(x,\tau) = \cosh\tau\sinh x\tau - x\sinh\tau\cosh x\tau,$$

$$\beta = \frac{1-x^2}{8} + \frac{1-Q}{4\sinh^2\tau}, \qquad \psi = \frac{1-x^2}{8} + \frac{Q}{4\sinh^2\tau}, \qquad \gamma = \frac{R}{4\sinh^2\tau},$$

$$E_N(\tau) = k\coth\tau - \frac{k^2}{2} - \frac{1}{2N^2}, \qquad k = \frac{\coth(\tau/N)}{N}.$$

**Theorem 3.3** (exact symbol; **[Lean]**). *For every $N \geq 2$, $1 \leq m \leq N-1$ and $\xi \in (0,1)$,*

$$\frac{a_m}{N^2} = E_N(\tau) - \frac{1}{4\sinh^2\tau} - \psi(x,\tau), \qquad \frac{b_m}{N^2} = \beta(x,\tau), \qquad \frac{g_m}{N^2} = \gamma(x,\tau).$$

*Lean:* `aBlock_xiOf`, `bBlock_xiOf`, `gBlock_xiOf`.

*Proof.* With $\xi^j = e^{-2j\tau/N}$: $\xi^m = e^{-(1-x)\tau}$, $1 - X = 2e^{-\tau}\sinh\tau$, and from the closed form of $W_k$, $(W_m + W_{N-m})/N = Q/(2\sinh^2\tau)$, $(W_m - W_{N-m})/N = 2\gamma$, $NW_N = N^2/(4\sinh^2\tau)$ (use $\sinh\tau + e^{-\tau} = \cosh\tau$). For $a_m$ note that the one-body part equals $Nc\coth\tau - \frac{c^2}{2} - \frac{1}{2}$ and $m(N-m) = N^2(1-x^2)/4$. □

The mode number enters only through $x$, and in $a_m$ only through $\psi$: up to an $m$-independent term the blocks are a function of $m/N$. This is what makes all $N$ tractable at once.

**Lemma 3.4** (tangential stiffness; **[Lean]**). *$\beta(x,\tau) > 0$ for $|x| < 1$, $\tau > 0$; hence $b_m > 0$ for every $N \geq 2$, $1 \leq m \leq N-1$, $\xi \in (0,1)$. Lean:* `beta_pos`, `bBlock_pos`.

*Proof.* With $a = (1+x)\tau$, $b = (1-x)\tau$ and $f(t) = \sinh^2(t/2)/t$ one checks $\beta = \frac{ab}{4\tau\sinh^2\tau}[f(a+b) - f(a) - f(b)]$. Since $f(t)/t$ is strictly increasing, $f$ is strictly superadditive on $(0,\infty)$. □

**Lemma 3.5** (critical mode; **[Lean]**). *For every $\tau > 0$, $\psi(\cdot,\tau)$ is even and strictly decreasing on $[0,1]$. Hence $a_m \geq a_{\lfloor N/2\rfloor}$, with equality only for $m \in \{\lfloor N/2\rfloor, \lceil N/2\rceil\}$, and the ring is spectrally stable iff $a_{\lfloor N/2\rfloor} \geq 0$; if $a_{\lfloor N/2\rfloor} < 0$ the modes $\lfloor N/2\rfloor, \lceil N/2\rceil$ grow exponentially. Lean:* `psi_strictAntiOn`, `aBlock_half_le`, `aBlock_half_lt`, `specStable_iff`.

The proof (a careful but elementary estimate of $\partial_x\psi$ through $\lambda(t) = \log(\sinh t/t)$) is in Appendix B.

### 3.2 One threshold for every $N \geq 8$

**Theorem 3.6** ([**Lean**]). *(a) For $2 \leq N \leq 7$ and every $\xi \in (0,1)$: $a_m > 0$, $b_m > 0$ and $a_m b_m - g_m^2 > 0$ for all $m$. The ring is spectrally stable and definite on $V$ at every radius.*

*(b) For every $N \geq 8$ there is $\xi_*(N) \in (0,1)$ such that $a_{\lfloor N/2 \rfloor} < 0$ on $(0, \xi_*)$, $= 0$ at $\xi_*$ and $> 0$ on $(\xi_*, 1)$. The ring is spectrally unstable for $\xi < \xi_*$ (in the modes $\lfloor N/2 \rfloor$, $\lceil N/2 \rceil$) and spectrally stable for $\xi \geq \xi_*$.*

*(c) $\xi_*(N)$ is algebraic: it is a root of the integer polynomial $2(1-\xi)^2(1-\xi^N)^2 a_{\lfloor N/2 \rfloor}$.*

*Lean: (a)* `stable_N2`, ..., `stable_N7`, `stable_all_radii_of_le_seven`; *(b)* `unique_threshold_of_ge_eight`; *(c) is immediate from Theorem 3.1.*

*Proof.* Spectral stability is equivalent to $a_{\lfloor N/2 \rfloor} \geq 0$ (Lemma 3.5). The proof of (b) splits by parity.

*Even $N$.* At $x = 0$ and with $\eta = \tau/2$, $\mu = \frac{2}{N} \coth \frac{2\eta}{N}$, Theorem 3.3 gives

$$\frac{8 a_{N/2}}{N^2} = 2\mu \tanh \eta - (\coth \eta - \mu)^2 - \frac{4}{N^2}. \tag{3}$$

For $N \geq 4$ this is strictly decreasing in $\eta$: $\mu \tanh \eta = \frac{2}{N} \tanh(qv)/\tanh v$ with $v = 2\eta/N$, $q = N/2 > 1$, has logarithmic derivative $\frac{2q}{\sinh 2qv} - \frac{2}{\sinh 2v} < 0$, and $\coth \eta - \mu \geq 0$ is non-decreasing, both because $t/\sinh t$ decreases. The limits are $2 - 4/N^2$ as $r \to \infty$ and the planar value $-(N^2 - 8N + 8)/N^2$ as $r \to 0$, which is negative iff $N \geq 8$ (Lean: `aBlock_half_threshold`, `aBlock_half_pos_of_le_six`).

*Odd $N \geq 17$.* Here $a_{(N-1)/2}/N^2 = \Phi_N(\tau) + \varepsilon_N(\tau)$, where $\Phi_N$ is the right side of (3) divided by 8 (strictly decreasing by the same argument) and $\varepsilon_N = \psi(0,\tau) - \psi(1/N, \tau) \in [0, 1/(4N^2)]$. Explicit bounds on $\Phi_N$, $\varepsilon_N$ and their $\tau$-derivatives show that $a_{(N-1)/2}$ is positive for $\tau \leq 5$, negative for $\tau \geq 10$ and strictly decreasing on $[5, 10]$ (Lean: `aBlock_half_threshold_of_ge_17`; the estimates are in `Tanh.LargeN.OddN`).

*Odd $9 \leq N \leq 15$ and all $N \leq 7$.* Exact computation: for each $N$ and $m$ the cleared numerators of $a_m$, $b_m$ and $a_m b_m - g_m^2$ are explicit integer polynomials, and their sign pattern on $(0,1)$ is proved in Lean by polynomial positivity certificates and a rational isolating interval for the root of $a_{\lfloor N/2 \rfloor}$ (`cert_N9`, `cert_N11`, `cert_N13`, `cert_N15`, `stable_N2`–`stable_N7`; independently by exact real-root isolation in `code/tanh_threshold_polys.py`). □

*Remark* 3.7 (algebraic structure; [**Certified**]/[**Lean**]). That $\xi_*(N)$ is algebraic is automatic from Theorem 3.1; the structure is not. For $8 \leq N \leq 16$ the minimal polynomial is palindromic (so $1/\xi_*$ is a Galois conjugate, the image of the ring under inversion in the boundary circle) of degree $2N - 2$ for odd $N$, $\frac{3}{2}N - 2$ for $N \equiv 0$, and $\frac{3}{2}N - 3$ for $N \equiv 2 \pmod 4$; $\xi_*(8)$ is a unit with minimal polynomial $x^{10} - 12x^9 - 37x^8 - 72x^7 - 84x^6 - 88x^5 - 84x^4 - 72x^3 - 37x^2 - 12x + 1$, while for $9 \leq N \leq 16$ the leading coefficient exceeds 1 and $\xi_*(N)$ is not an algebraic integer. Irreducibility is in Lean; the degree pattern for general $N$ is open.

*Remark* 3.8 (comparison with Boatto; [**Numerical**]). Boatto [2] fixes the geodesic radius $r_0 = 1$ and varies the curvature $K_{\mathrm{B}} < 0$; then $\xi = \tanh^2(\sqrt{-K_{\mathrm{B}}}/2)$. Implementing her Hessian (her eqs. (14), (17), (29), (31); equation numbers as in the journal version) independently of our blocks, the zero crossing of her determinant for $N = 8$ is $K_{\mathrm{B}} = -0.28218454857650183188$, which maps to $\xi_*(8)$ to $10^{-41}$; for $N = 10$ the agreement is $10^{-42}$. For $N = 9$ her (definiteness) criterion gives $\xi_{\mathrm{EM}}(9)$, i.e. $\xi_*(9) + 3.22 \cdot 10^{-8}$, as it must by Theorem 4.2. Her mode blocks are ours up to $m$-independent positive factors (`code/boatto_map_and_verify.py`).

| $N$ | $\xi_*(N)$ | $r_*(N)$ | $\cosh r_*(N)$ | deg | $\xi_{\mathrm{EM}} - \xi_*$ |
|---|---|---|---|---|---|
| 8 | 0.06735622300408126226843 74… | 0.5312104560 | 1.1444414784 | 10 | 0 |
| 9 | 0.11287106323660274043969 40… | 0.6990695396 | 1.2544637167 | 16 | $3.220 \cdot 10^{-8}$ |
| 10 | 0.16611424152623911808175 86… | 0.8653901256 | 1.3984100696 | 12 | 0 |
| 11 | 0.20460117688266929063553 58… | 0.9752480466 | 1.5144618547 | 20 | $1.244 \cdot 10^{-7}$ |
| 12 | 0.24741777680406770917396 50… | 1.0917203545 | 1.6575169309 | 16 | 0 |
| 13 | 0.27986138864390668702999 65… | 1.1775646302 | 1.7772431147 | 24 | $1.777 \cdot 10^{-7}$ |
| 14 | 0.31493896311550662339350 54… | 1.2691494599 | 1.9194478920 | 18 | 0 |
| 15 | 0.34245858351217025660102 26… | 1.3406681986 | 2.0416335000 | 28 | $1.886 \cdot 10^{-7}$ |
| 16 | 0.37168181390604545663672 87… | 1.4167509599 | 2.1831006077 | 22 | 0 |

Table 1: Ring thresholds, $8 \le N \le 16$. **[Certified]**: $\xi_*(N)$ lies in an exact rational interval of width $< 10^{-45}$ on which the cleared critical numerator changes sign, and in an Arb ball of radius $< 3{\cdot}10^{-153}$ (`code/tanh_threshold_polys.py`); Lean locates it in a rational interval of width $10^{-20}$ (`cert_N8`, …, `cert_N16`). $r_*$ and $\cosh r_*$ are evaluated from $\xi_*$. "deg" is the degree of the minimal polynomial of $\xi_*$: the factor $f_N$ of the critical numerator carrying $\xi_*$ is irreducible over $\mathbb{Q}$ (Lean: `f8_irreducible`, …, `f16_irreducible`, by mod-$p$ certificates), and palindromic. Last column: the Krein window (Theorem 4.2; **[Certified]**, `code/tanh_krein.py`).

# 4 Odd $N$: the Krein window

Let $\xi_{\mathrm{EM}}(N) = \inf\{\xi_0 : \ d^2F \text{ is definite on } V \text{ for all } \xi \in (\xi_0, 1)\}$. Since definiteness implies spectral stability, $\xi_{\mathrm{EM}} \ge \xi_*$.

**Lemma 4.1** (definiteness excess; **[Lean]**). *Let $\Phi_N(\tau) = E_N(\tau) - \frac{1}{4\sinh^2\tau} - \psi(0,\tau)$ and $\mathcal{G} = \psi(0,\tau) - \psi(x,\tau) - \gamma^2/\beta$. Then $\mathcal{G} \ge 0$, with equality only at $x = 0$, and $a_m b_m - g_m^2 = N^4\beta\,[\Phi_N(\tau) + \mathcal{G}(x,\tau)]$. Lean: `beta_mul_G`, `K_pos`, `G_nonneg`, `det_xiOf`.*

*Proof.* With $C = \cosh\tau$, $S = \sinh^2\tau$: $8SB\mathcal{G} = AB - 4R^2$, where $A = x^2S + 2C - 2Q$ and $B = 8S\beta$; using $Q^2 - R^2 = C^2 - x^2S$ this equals $2(C+1)^2\mathcal{K}$ with $\mathcal{K} = C(1-\cosh x\tau) + x\sinh\tau\sinh x\tau - \frac{x^2(C-1)}{2}[2-(1-x^2)(C-1)]$. Now $\mathcal{K}(x,0) = 0$ and $\partial_\tau\mathcal{K} = \sinh\tau(1-x^2)[x^2(C-1) - (\cosh x\tau - 1)] > 0$ for $0 < |x| < 1$. □

**Theorem 4.2** (Krein window). *(a) **[Lean]** For even $N$ and every radius, $d^2F$ is definite on $V$ iff $a_{N/2} > 0$; hence $\xi_{\mathrm{EM}}(N) = \xi_*(N)$ and there is no window. (`emDefinite_iff_even`, `xiEM_eq_xiStar_even`.)*

*(b) **[Lean]** For every odd $N \ge 9$: $0 < \xi_* < \xi_{\mathrm{EM}} < 1$; the ring is spectrally stable on $(\xi_*, 1)$; and there is $\eta > 0$ with $a_m b_m - g_m^2 < 0$ for $m = \frac{N-1}{2}$ on $(\xi_*, \xi_* + \eta)$. Moreover $\xi_{\mathrm{EM}} - \xi_* = O(N^{-3})$. (`krein_window_odd`, `HyperbolicRings.Krein.krein_window_bounds`, `xiEM_lt_one`, `xiEM_sub_xiStar_le`.)*

*(c) **[Lean]** On that interval the critical mode has two distinct purely imaginary eigenvalues $\rho_0\lambda_\pm = i(-g_m \pm \sqrt{a_m b_m})$, and the quadratic form $d^2F$ has opposite signs on their eigenvectors (opposite Krein signatures). (`HyperbolicRings.Krein.ring_window_hypotheses`, `opposite_signatures`, `eigenvalues_imaginary_distinct`.)*

*(d) **[Certified]** For $N = 9, 11, 13, 15$ the window $(\xi_*, \xi_{\mathrm{EM}})$ is given in Table 1 (disjoint exact rational isolating intervals), and on the whole window the critical block is indefinite while every other block is positive definite (`code/tanh_krein.py`). For $N \in \{17, 21, 31, 61, 121, 241\}$ the same holds at sampled points **[Numerical]** (`code/tanh_extras.py`).*

*Proof.* (a) By Lemma 3.5, $a_m \geq a_{N/2} = N^2\Phi_N$; if $\Phi_N > 0$ Lemma 4.1 gives $a_m b_m - g_m^2 > 0$ for all $m$.

(b) The planar limit $-(N-1)(N-7)/N^2 < 0$ of $8a_{(N-1)/2}/N^2$ gives $\xi_* > 0$. At $\xi_*$, $a_m b_m - g_m^2 = -g_m^2 < 0$ because $R(1/N, \tau) > 0$ (tanh is strictly concave on $[0, \infty)$, so $\tanh x\tau > x \tanh \tau$); continuity gives $\eta$. For the width: $\Phi_N > 0$ implies definiteness (Lemma 4.1), $\Phi_N$ has slope bounded away from 0 near $\tau_\infty$ (§5), and $a_{(N-1)/2}/N^2 - \Phi_N = \varepsilon_N = O(N^{-2})$; so the two $\tau$-thresholds differ by $O(N^{-2})$, and $\xi = e^{-2\tau/N}$ gives a further factor $2/N$.

(c) $a_m b_m > 0$ on the interval, since $a_m > 0$ above $\xi_*$ (Theorem 3.6) and $b_m > 0$. For $M = \left(\begin{smallmatrix} a & ig \\ -ig & b \end{smallmatrix}\right)$ and $\mu = \pm\sqrt{ab}$, every eigenvector of $J_0 M$ for $i(-g + \mu)$ is a multiple of $(b, i\mu)$, and $(b, i\mu)^* M (b, i\mu) = 2b\mu(\mu - g)$. The product of the two values is $4b^2\, ab\, (ab - g^2) < 0$. □

**What the window means.** Throughout $(\xi_*, \xi_{\mathrm{EM}})$ the odd ring is spectrally stable, with semisimple spectrum on $V$ (the rotation mode $m = 0$ carries a Jordan block whenever $c_0 \neq 0$; numerically $c_0 > 0$ for every $N$ checked), but no Lyapunov function of energy–momentum type exists, and the two critical frequencies, of opposite signature, may collide under a perturbation (for example, a change of the model) and produce a Hamiltonian–Hopf instability. Whether the ring is nonlinearly stable inside the window is open. A criterion based on definiteness of the Hessian, such as the one used in [2], reports the threshold $\xi_{\mathrm{EM}}$ and so misses the window; for $N \leq 15$ the difference is below $2 \cdot 10^{-7}$ in $\xi$. Expanding the critical block suggests $N^3(\xi_{\mathrm{EM}} - \xi_*) \to 0.003709\ldots$ **[Derived]** (assuming the critical block is the last to become definite); the data give 0.00348 at $N = 241$.

# 5 Large $N$ and the horocycle row

## 5.1 The limit function

As $N \to \infty$ at fixed $\tau$, $E_N \to \coth\tau/\tau - 1/(2\tau^2)$ and the critical block tends to $a_{\lfloor N/2 \rfloor}/N^2 \to \Phi(\tau)$, with

$$8\,\Phi(\tau) = \frac{4}{\tau}\tanh\frac{\tau}{2} - \mathcal{L}\Big(\frac{\tau}{2}\Big)^2, \qquad \mathcal{L}(\eta) = \coth\eta - \frac{1}{\eta}. \tag{4}$$

**Lemma 5.1** (**[Lean]**; digits **[Certified]**)**.** *$\Phi$ is strictly decreasing on $(0, \infty)$ from $\frac{1}{4}$ to $-\frac{1}{8}$; its unique zero $\tau_\infty$ satisfies $7.4453 < \tau_\infty < 7.4454$ (Lean: `Phi_strictAntiOn`, `existsUnique_Phi_zero`, `tauInf_bracket`). An Arb bracket gives*

$$\tau_\infty \in (7.4453392449490186871,\ 7.4453392449490186873),$$

*so $\ell_* := 2\pi/\tau_\infty = 0.84390853129253347\ldots$ and $\Phi'(\tau_\infty) = -0.01532424264\ldots$ (`code/tanh_large_n.py`).*

*Proof.* $(\tanh\eta/\eta)' < 0$ and $(\mathcal{L}^2)' = 2\mathcal{L}\mathcal{L}' > 0$. The bracket is a sign evaluation with explicit bounds for exp. □

Dropping the exponentially small terms turns $\Phi(\tau_\infty) = 0$ into $\eta^2 - 4\eta + 1 = 0$ for $\eta = \tau/2$; so $\tau_\infty/2 \approx 2 + \sqrt{3}$ and $\ell_* \approx \pi(2 - \sqrt{3}) = 0.8418$ **[Numerical]**.

**Theorem 5.2** (large-$N$ threshold; **[Lean]**)**.** *For both parities of $N$,*

$$\cosh r_*(N) = \frac{N}{\tau_\infty} + O\Big(\frac{1}{N}\Big), \qquad \ell_N := \frac{2\pi\sinh r_*(N)}{N} = \ell_* + O(N^{-2}),$$

*and the same holds for $\xi_{\mathrm{EM}}(N)$. Lean: `cR_xiStar_rate`, `tendsto_cR_xiStar`, `spacing_rate`, `cR_xiEM_rate`.*

*Proof.* By Theorem 3.3, $a_{\lfloor N/2\rfloor}/N^2 = \Phi(\tau) + \Delta_N(\tau) + \varepsilon_N(\tau)$ with $\Delta_N = E_N - E_\infty$ and $\varepsilon_N = \psi(0,\tau) - \psi(x_N,\tau)$, $x_N \in \{0, 1/N\}$. Writing $\delta = \mathcal{L}(\tau/N)/N$ one finds $-\frac{1}{2N^2} - \frac{\delta^2}{2} \le \Delta_N \le \frac{\tau}{3N^2}$, and $0 \le \varepsilon_N \le 1/(4N^2)$. Since $\Phi'(\tau_\infty) \neq 0$, $a_{\lfloor N/2\rfloor}$ is positive at $\tau_\infty - K/N^2$ and negative at $\tau_\infty + K/N^2$ for large $K$, so $|\tau_*(N) - \tau_\infty| \le K/N^2$ by uniqueness of the threshold. Finally $\cosh r_* = \coth(\tau_*/N)$ and $\sinh r_* = 1/\sinh(\tau_*/N)$. □

So at threshold the neighbours of a large ring are $\ell_* \approx 0.844$ curvature radii apart along the ring (geodesic distance $2\,\mathrm{arsinh}(\ell_*/2) = 0.82068\ldots$). In the sinh model, by contrast, $\cosh r_* \sim \sqrt{N}/2$ and the threshold spacing tends to 0 [11].

**Proposition 5.3** (second order; **[Proved]**, constants **[Numerical]**). *For each parity there is $N_0$ such that for $N \ge N_0$ the quantities $\tau_*(N)$, $\cosh r_*(N)/N$ and $\ell_N$ are values at $N^{-2}$ of real-analytic functions, and*

$$\cosh r_*(N) = \frac{N}{\tau_\infty} + \frac{\varsigma}{N} + O(N^{-3}), \qquad \ell_N = \ell_* + \frac{2\pi(\varsigma - \tau_\infty/2)}{N^2} + O(N^{-4}),$$

$$\varsigma = \frac{\tau_\infty}{3} - \frac{\mathcal{D} + [N \ \mathit{odd}]\, p(\tau_\infty)}{|\Phi'(\tau_\infty)|\,\tau_\infty^2}, \qquad \mathcal{D} = \frac{\tau_\infty \coth\tau_\infty - 1}{3} - \frac{1}{2}, \qquad p(\tau) = -\tfrac{1}{2}\,\partial_x^2\psi(0,\tau).$$

*Numerically* $\varsigma_{\mathrm{e}} = 0.5412188400$, $\varsigma_{\mathrm{o}} = 0.4010382403$, $\ell_N - \ell_* \approx -19.990N^{-2}$ *(even) and* $-20.870N^{-2}$ *(odd); the thresholds give* $N(\cosh r_* - N/\tau_\infty) = 0.5412206$ *at* $N = 800$ *and* $0.4010393$ *at* $N = 801$ *(`code/b4_varsigma.py`, `code/tanh_large_n.py`; Figure 2).*

*Proof.* With $\epsilon = N^{-2}$ and $y = \tau/N$, $k = (y\coth y)/\tau$ is an even analytic function of $y$, hence analytic in $(\tau, \epsilon)$ ($y^2 = \tau^2\epsilon$), and $\psi(x,\tau)$ is even in $x$, so $\psi(1/N,\tau)$ is analytic in $\epsilon$. Thus $a_{\lfloor N/2\rfloor}/N^2 = \Psi_{\mathrm{par}}(\tau,\epsilon)$ with $\Psi_{\mathrm{par}}$ analytic near $(\tau_\infty, 0)$ and $\Psi_{\mathrm{par}}(\tau,0) = \Phi(\tau)$. Since $\Phi'(\tau_\infty) \neq 0$ the implicit function theorem gives an analytic zero $\tau(\epsilon)$, which is $\tau_*(N)$ by Theorem 5.2 and uniqueness. Expanding $y\coth y = 1 + y^2/3 + \ldots$ gives $\partial_\epsilon\Psi_{\mathrm{par}}(\tau,0) = \mathcal{D}(\tau) + [N \text{ odd}]p(\tau)$, so $\tau_* = \tau_\infty + \epsilon(\mathcal{D} + [\mathrm{odd}]p)/|\Phi'| + O(\epsilon^2)$; then $\cosh r_* = \coth(\tau_*/N) = N/\tau_* + \tau_*/(3N) + O(N^{-3})$ and $\ell_N = 2\pi/(N\sinh(\tau_*/N)) = 2\pi/\tau_* - 2\pi\tau_*/(6N^2) + O(N^{-4})$. □

Lean computes $\partial_x^2\psi(0,\tau) = -\frac{1}{4} + \frac{\tau^2\cosh\tau - 2\tau\sinh\tau}{4\sinh^2\tau}$ and proves that it is negative at $\tau_\infty$, so $p(\tau_\infty) > 0$ and $\varsigma_{\mathrm{o}} < \varsigma_{\mathrm{e}}$ (`HyperbolicRings.Horocycle.second_deriv_psi`, `psiXX_tauInf_neg`); the implicit-function step is not in Lean.

## 5.2 The horocycle row

In the upper half-plane (metric $|dz|^2/y^2$) the pair term is $-\log\tanh(d/2) = -\log|z-w| + \log|z-\bar w|$: a planar row plus its mirror image, with the hyperbolic symplectic form $dx \wedge dy/y^2$. Take the horocycle $y = 1$ and the row $z_k = k\ell + i$, $k \in \mathbb{Z}$. The translations $x \mapsto x + t$ are isometries with momentum $J = -\sum_k 1/y_k$. The forces on a vortex converge absolutely (the pair and image terms together decay like $k^{-3}$ and $k^{-2}$), and the Hessian couples sites through absolutely summable $2\times 2$ blocks, so on a Bloch wave $(\delta x_k, \delta y_k) = e^{2\pi i\theta k}(\delta x, \delta y)$ it acts by a Hermitian matrix $M(\theta)$.

**Theorem 5.4** (horocycle row; **[Lean]**). *Let $\tau = 2\pi/\ell$ and $x = 1 - 2\theta$.*

*(a) The row is a relative equilibrium drifting at $\dot x = \frac{1}{2}(\tau\coth\tau - 1)$, and, in the basis $(\delta x, \delta y)$,*

$$\Big(\frac{\ell}{2\pi}\Big)^2 M(\theta) = \begin{pmatrix} \beta & i\gamma \\ -i\gamma & \mathfrak{a} \end{pmatrix}(x,\tau), \qquad \mathfrak{a} = \frac{\coth\tau}{\tau} - \frac{1}{2\tau^2} - \frac{1}{4\sinh^2\tau} - \psi(x,\tau),$$

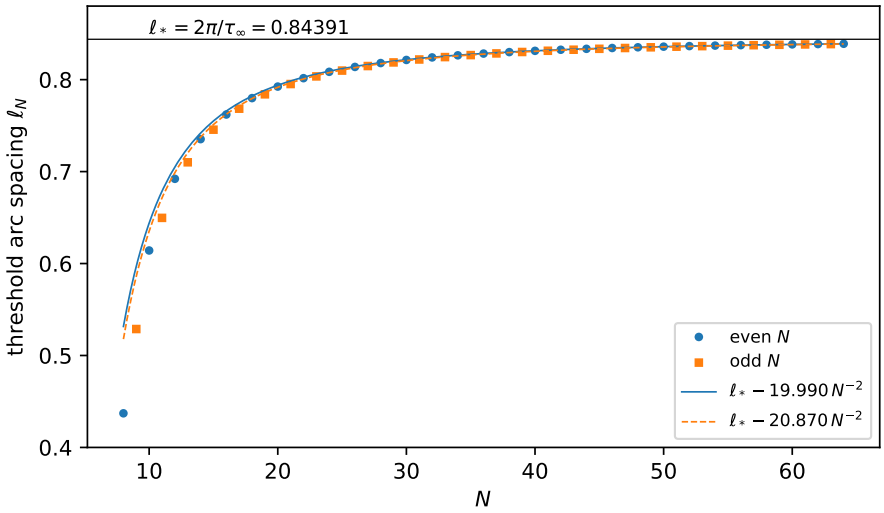



Figure 2: Threshold arc spacing $\ell_N = 2\pi \sinh r_*(N)/N$ of the $N$-ring against $N$, with the limit $\ell_*$ (Theorem 5.2) and the second-order curves of Proposition 5.3 for each parity. Floating point.

*with $\beta, \gamma, \psi$ as in Theorem 3.3. So the ring blocks divided by $N^2$ converge, at fixed $(m/N, \tau)$, to the row symbol, with ring mode number $m/N$ becoming the Bloch number $\theta$ and the ring's (tangential, outward radial) directions becoming $(\delta x, -\delta y)$. Note the order of the basis: the ring blocks $M_m$ are written in the order (radial, tangential), so their $(1,1)$ entry is the radial stiffness $a_m$, while $M(\theta)$ is written in the order $(\delta x, \delta y)$ = (tangential, radial), so its $(1,1)$ entry $\beta$ is the tangential stiffness and its $(2,2)$ entry $\mathfrak{a}$ is the limit of $a_m/N^2$.*

*(b) Every Bloch exponent is imaginary iff $\ell \geq \ell_*$; $M(\theta)$ is positive definite for all $0 < \theta < 1$ iff $\ell > \ell_*$, and at $\ell = \ell_*$ it degenerates exactly at the pairing mode $\theta = \frac{1}{2}$. There is no Krein window.*

*Lean: (a)* `row_bloch_symbol`*, from the pair Hessian of* `pair_hessian_row` *and the four lattice sums of* `Tanh.LargeN.RowFourier` *(the last by Poisson summation); (b)* `row_stable_iff`*,* `row_definite_iff`*,* `row_spectral_iff`*.*

*Proof sketch.* (a) With $p_k = (k\ell - 2i)^{-2}$ the coupling blocks are $\mathrm{diag}(-1,1)(k\ell)^{-2} + \begin{pmatrix} \mathrm{Re}\,p_k & \mathrm{Im}\,p_k \\ -\mathrm{Im}\,p_k & \mathrm{Re}\,p_k \end{pmatrix}$; the sums $\sum_{k\neq 0} k^{-2}$, $\sum \cos(qk)/k^2$, $\sum (k^2+a^2)^{-1} = \frac{\pi}{a}\coth \pi a$ and $\sum_k e^{iqk}(k-ia)^{-2} = -2\pi \sum_{n\geq 0}(q+2\pi n)e^{-a(q+2\pi n)}$ evaluate every entry, and the last one reproduces the quantity $W_m/N$ of Theorem 3.3. (b) $M_{xx} \propto \beta > 0$ (Lemma 3.4) and $M_{yy} \propto \mathfrak{a} = \Phi(\tau) + \psi(0,\tau) - \psi(x,\tau) \geq \Phi(\tau)$ with equality at $\theta = \frac{1}{2}$ (Lemma 3.5); so stability is $\Phi(2\pi/\ell) \geq 0$, i.e. $\ell \geq \ell_*$. Definiteness follows from $\mathfrak{a}\beta - \gamma^2 = \beta(\Phi + \mathcal{G})$ (Lemma 4.1). □

Theorem 5.2 therefore says: a large ring loses stability exactly where the horocycle row with the same spacing does, and in the same (pairing) mode. As $\ell \to 0$, $8\Phi \to -1$: a dense horocycle row is unstable, like the planar row; as $\ell \to \infty$, $8\Phi \to 2$. Spectral stability of the row is not bounded linear stability: at $\theta = 0$ there is a Jordan block for every $\ell$ (a uniform change of height makes the row drift). We make no nonlinear claim for infinite rows.

# 6 Rows on hypercycles

## 6.1 Setting and Bloch symbol

In the upper half-plane write $z = e^{u+i\varphi}$, $u \in \mathbb{R}$, $\varphi \in (0,\pi)$ (the "strip chart"). The metric is $|du + i\,d\varphi|^2/\sin^2\varphi$, the curve $\varphi = \alpha$ is a hypercycle (equidistant from the geodesic $\varphi = \pi/2$) of geodesic curvature $\kappa = \cos\alpha$, $u$-translations are isometries (hyperbolic boosts), and

$$X_{ij} := \cosh d_{ij} = 1 + \frac{\cosh(u_i - u_j) - \cos(\varphi_i - \varphi_j)}{\sin\varphi_i \sin\varphi_j}, \qquad W(X) = -\log\tanh\frac{d}{2} = \frac{1}{2}\log\frac{X+1}{X-1}.$$

In this chart the pair term is

$$W = \tfrac{1}{2}\big[\log(\cosh(u_i - u_j) - \cos(\varphi_i + \varphi_j)) - \log(\cosh(u_i - u_j) - \cos(\varphi_i - \varphi_j))\big], \tag{5}$$

which is $-\log|z_i - z_j| + \log|z_i - \bar{z}_j|$ at $z = e^{u+i\varphi}$ ([**Lean**]: `HyperbolicRings.Rows.stripW_eq_pairG`; the same upper-half-plane pair term as for the horocycle row). The area form is $du\,d\varphi/\sin^2\varphi$, so Hamilton's equations are $\dot{u}_i = \sin^2\varphi_i\,\partial_{\varphi_i}H$, $\dot{\varphi}_i = -\sin^2\varphi_i\,\partial_{u_i}H$; the momentum of the $u$-translations, $\sum_i \cot\varphi_i$, adds only a constant drift to $\dot{u}$ and nothing to the linearisation. The row $u_k = k\ell_u$, $\varphi_k = \alpha$ has arc spacing $\ell = \ell_u/\sin\alpha$ and is a relative equilibrium: by symmetry $\dot{\varphi}_k = 0$ and all vortices drift with the same $\dot{u}$. The reflection $\varphi \mapsto \pi - \varphi$ maps $\kappa$ to $-\kappa$; we take $\kappa \geq 0$, $K := \kappa^2 \in [0,1)$, $\sigma := 1 - K = \sin^2\alpha$. Put $c_j = \cosh(j\ell_u)$ and $D_j = c_j + 1 - 2K$.

**Proposition 6.1** (Bloch symbol; [**Lean**])**.** *On the Bloch wave $(\delta u_k, \delta\varphi_k) = e^{iqk}(\delta u, \delta\varphi)$ the linearisation in the co-moving frame acts as $\left(\begin{smallmatrix} i\varpi(q) & B(q) \\ C(q) & i\varpi(q)\end{smallmatrix}\right)$, with real $\varpi$ and*

$$B(q) = 2\sum_{j\geq 1}\big[B^{(0)}(c_j) + B^{(1)}(c_j)\cos qj\big], \qquad C(q) = 2\sum_{j\geq 1} C^{(1)}(c_j)(1 - \cos qj),$$

$$B^{(0)} = \sigma\,\frac{(3K-1)c^2 - (4K^2 - K + 1)c + 2K^2}{(c-1)D^2}, \qquad B^{(1)} = \sigma\,\frac{Kc^2 + (1-3K)c + 2K^2 - 2K + 1}{(c-1)D^2},$$

$$C^{(1)} = -\sigma^2\,\frac{c^2 + c - 2K}{(c-1)D^2}.$$

*The exponents are $i\varpi \pm \sqrt{B(q)C(q)}$. Since $C^{(1)} < 0$, $C(q) < 0$ for $q \neq 0$, and the row is spectrally stable iff*

$$\mathcal{F}(q) := \tfrac{1}{2}B(q) = \sum_{j\geq 1}\big[B^{(0)}(c_j) + B^{(1)}(c_j)\cos qj\big] \;\geq\; 0 \qquad \text{for all } q \in (0,\pi]. \tag{6}$$

*Lean: the eight per-pair coefficients of the linearised velocity of vortex 0 (second partial derivatives of (5) at the row pair, as iterated one-variable derivatives) are computed in closed form and identified with $B^{(0)}, B^{(1)}, C^{(1)}$ (`onB_eq_B0`, `cpB_eq_B1`, `onC_eq_C1`, `cpC_eq_C1`, `onU_val`, …, `cpP_val`); their Bloch lattice sums over $k \neq 0$ converge and give the four entries (`hasSum_symB`, `hasSum_symC`, `hasSum_symU`, `hasSum_symP`), $C(q) < 0$ for $\cos q \neq 1$ (`Cser_neg`), and the mode is spectrally stable iff $\mathcal{F}(q) \geq 0$ (`bloch_stable_iff`).*

Lean does not derive Hamilton's equations in the strip chart from those in the upper half-plane (the factor $\sin^2\varphi$ is the conformal factor of the chart), and, as for the horocycle row, it identifies the linearisation of an infinite row with the lattice sum of per-pair second derivatives rather than differentiating an infinite sum. The derivation by hand is in Appendix C; for a general kernel $W(X)$

the pair functions are $B^{(0)} = cW' + K(X-1)^2W''$, $B^{(1)} = -(1-K(X-1))W' + K(X-1)^2W''$, $C^{(1)} = -cW' - (c+1)(X-1)W''$ at $X = 1 + (c-1)/\sigma$. The symbol is also checked against a direct numerical Hessian (40 digits) and against the spectrum of a periodic 6-site supercell, to $10^{-29}$ (`code/rows_symbol.py`).

**Lemma 6.2** (pair identities; **[Lean]**). *For $0 \le K < 1$ and $c > 1$, with $\mathfrak{g}_{\mathrm{o}} = \frac{(2K-1)c-1}{(c-1)D}$ and $\mathfrak{g}_{\mathrm{e}} = \frac{(4K-1)c-(4K^2-2K+1)}{D^2}$:*

$$B^{(0)} - B^{(1)} = \sigma\,\mathfrak{g}_{\mathrm{o}}, \qquad B^{(0)} + B^{(1)} = \sigma\,\mathfrak{g}_{\mathrm{e}}, \qquad B^{(1)} > 0, \qquad C^{(1)} < 0.$$

*For $K \le \frac{1}{2}$: $\mathfrak{g}_{\mathrm{o}} \le -\frac{1}{(c-1)c}$ and $\mathfrak{g}_{\mathrm{e}} \le \frac{c-1}{c^2}$ (the values at $K = \frac{1}{2}$). Lean: `HyperbolicRings.Rows.B0_sub_B1`, `B0_add_B1`, `B1_pos`, `C1_neg`, `go_le_half`, `ge_le_half`.*

*Proof.* The two identities are rational-function identities. In $t = c-1$ the numerator of $B^{(1)}$ is $Kt^2 + (1-K)t + 2(1-K)^2 > 0$. With $T = 1-2K \ge 0$: $-\frac{1}{(c-1)c} - \mathfrak{g}_{\mathrm{o}} = \frac{T(c+1)}{c(c+T)} \ge 0$ and $\frac{c-1}{c^2} - \mathfrak{g}_{\mathrm{e}} = \frac{T(2c^3+c^2+Tc^2+Tc-2c-T)}{c^2(c+T)^2} \ge 0$. □

**Proposition 6.3** (critical mode; **[Lean]**, eq. (7) **[Proved]**). *$\mathcal{F}$ is strictly decreasing on $[0,\pi]$ and $\mathcal{F}(q) \ge \mathcal{F}(\pi)$ for every $q$. Hence the row is spectrally stable iff $\mathcal{F}(\pi) \ge 0$, and the first mode to lose stability is the pairing mode $q = \pi$ (neighbours displaced in opposite directions). Summing the generating functions,*

$$\mathcal{F}(\pi) = \sum_{n\ge 1} \Big[ \big( \sin 2\alpha \sin 2n\alpha - 2n \sin^2\alpha \sin^2 n\alpha \big) \frac{\mathfrak{q}^n}{1-\mathfrak{q}^n} - 2n \sin^2\alpha \cos^2 n\alpha \, \frac{\mathfrak{q}^n}{1+\mathfrak{q}^n} \Big], \qquad \mathfrak{q} = e^{-\ell_u}. \tag{7}$$

*Lean: `hasSum_n_cos`, `hasSum_cos_geom`, `B1_gen` (the generating function of $B^{(1)}$), `F_strictAntiOn`, `F_ge_F_pi` (via $\cos((j+1)q) = T_{j+1}(\cos q)$), `row_stable_iff_F_pi`. Eq. (7) itself is not in Lean.*

*Proof.* With $z = e^{-x}$, $c = \cosh x$, the rational identity $B^{(1)}(c) = 2\sigma \sum_{n\ge1} n\cos^2(n\alpha) z^n$ holds (expand $\sum n\cos^2(n\alpha) z^n = \frac{1}{2}\frac{z}{(1-z)^2} + \frac{1}{4}\sum_{\pm} \frac{z e^{\pm 2i\alpha}}{(1-ze^{\pm 2i\alpha})^2}$; checked exactly in `code/kappa_c_check.py`). So $\sum_j B^{(1)}(c_j)\cos qj = 2\sigma \sum_n n\cos^2(n\alpha) P(r_n, q)$ with $r_n = e^{-n\ell_u}$ and $P(r,q) = \sum_{j\ge1} r^j \cos qj = \frac{r\cos q - r^2}{1-2r\cos q + r^2}$, and $\partial_q P = -\frac{r(1-r^2)\sin q}{(1-2r\cos q+r^2)^2} < 0$ on $(0,\pi)$; the coefficients $n\cos^2(n\alpha)$ are $\ge 0$ and not all zero (for $n=1$ or $n=2$). The interchange of the two sums is justified by absolute convergence ($\sum_{j,n} n e^{-n(j+1)\ell_u} < \infty$). For (7) use also $C^{(1)} = -2\sigma \sum_n n \sin^2(n\alpha) z^n$ (checked exactly in the same script), $B^{(0)} = C^{(1)} + \sin 2\alpha \sum_n \sin(2n\alpha) z^n$ (`code/rows_symbol.py`, part (a)), and sum over $j$ with signs $(-1)^j$. □

## 6.2 The critical curvature

**Theorem 6.4** (critical curvature $\kappa_c = 1/\sqrt{2}$; **[Lean]**). *Let $\ell_u > 0$.*

*(a) If $|\kappa| \le \frac{1}{2}$ (geodesics included), $\mathcal{F}(q) < 0$ for every $q$: every Bloch mode $q \ne 0$ is unstable.*

*(b) If $|\kappa| \le \kappa_c = 1/\sqrt{2}$, $\mathcal{F}(\pi) < 0$: the pairing mode has the real exponents $\pm\sqrt{B(\pi)C(\pi)}$, and the row is unstable at every spacing.*

*(c) If $|\kappa| > \kappa_c$ and $(2\kappa^2 - 1)\cosh\ell_u \ge 1$, then $\mathcal{F}(q) > 0$ for every $q$: the row is spectrally stable. In arc spacing the condition is $\ell \ge \ell_A(\kappa) := \operatorname{arcosh}\big(1/(2\kappa^2-1)\big)/\sqrt{1-\kappa^2}$.*

*So $\kappa_c = 1/\sqrt{2}$ is the exact critical curvature: no row with $|\kappa| \le \kappa_c$ is stable, and every row with $|\kappa| > \kappa_c$ is stable at large enough spacing. Lean, for the Bloch matrices of Proposition 6.1:* `HyperbolicRings.Rows.every_mode_unstable`, `row_unstable_of_le_kappa_c`, `row_stable_of_bound` *(from* `F_summable`, `all_mode_instability`, `pairing_instability`, `stability_bound`*).*

*Proof.* Since $B^{(1)} > 0$ and $|\cos| \le 1$, each term of (6) lies between $\sigma\,\mathfrak{g}_{\mathrm{o}}(c_j)$ and $\sigma\,\mathfrak{g}_{\mathrm{e}}(c_j)$ (Lemma 6.2); the series converges absolutely because $|\mathfrak{g}_{\mathrm{o,e}}(c)| \le 3\frac{c_1+1}{c_1-1}\frac{1}{c-1}$ for $c \ge c_1$ and $c_j - 1 \ge j^2\ell_u^2/2$.

(a) For $K \le \frac{1}{4}$ the numerator of $\mathfrak{g}_{\mathrm{e}}$ is $(4K-1)c - (4K^2-2K+1) < 0$, so every term is $\le \sigma\,\mathfrak{g}_{\mathrm{e}}(c_j) < 0$.

(b) At $q = \pi$, $\cos \pi j = (-1)^j$, so the $j$-th term is $\sigma\,\mathfrak{g}_{\mathrm{o}}(c_j)$ for odd $j$ and $\sigma\,\mathfrak{g}_{\mathrm{e}}(c_j)$ for even $j$. For $K \le \frac{1}{2}$, Lemma 6.2 bounds them by $-\frac{\sigma}{(c_j-1)c_j} < 0$ (odd) and $\sigma\frac{c_j-1}{c_j^2} < \frac{\sigma}{c_j} \le 2\sigma e^{-j\ell_u}$ (even). Keeping the first odd term and bounding all terms with $j \ge 2$ by $2\sigma e^{-j\ell_u}$,

$$\frac{\mathcal{F}(\pi)}{\sigma} \le -\frac{1}{(c_1-1)c_1} + \frac{2e^{-2\ell_u}}{1-e^{-\ell_u}}.$$

With $u = e^{\ell_u} > 1$, $(c_1-1)c_1 = \frac{(u-1)^2(u^2+1)}{4u^2}$ and the right side is negative iff $2u^3 > (u-1)(u^2+1)$, i.e. $u^3 + u^2 - u + 1 > 0$, which holds.

(c) If $(2K-1)c_1 \ge 1$ then $(2K-1)c_j \ge 1$ for all $j$, so $\mathfrak{g}_{\mathrm{o}}(c_j) \ge 0$, strictly for $j \ge 2$; every term is $\ge \sigma\,\mathfrak{g}_{\mathrm{o}}(c_j)$. □

The bound in (b) is simpler than the two-regime argument of the source draft [14]; `code/kappa_c_check.py` checks it on a grid.

*Remark* 6.5 (why 45°). At large spacing only nearest neighbours matter and $\mathcal{F}(\pi) = 2\sin^2\alpha\cos 2\alpha\, e^{-\ell_u} + O(e^{-2\ell_u})$; the sign is that of $\cos 2\alpha = 2\kappa^2 - 1$. The hypercycle $\varphi = \alpha$ meets the ideal boundary at the angle $\alpha$, so the critical one meets it at 45°. For any kernel decaying like $Ae^{-\gamma d}$ the same leading-order computation gives the same $\kappa_c$, whatever $A, \gamma$ [**Derived**]. For the screened kernels $W_\nu$ (Green functions of $\Delta - \nu(\nu+1)$, $\nu \ge 0$; $\nu = 0$ is ours) we only compute a few thresholds of $\nu = 1$ for orientation (`code/rows_screened.py`, [**Numerical**]) and do not treat them further.

### 6.3 One threshold curve

**Theorem 6.6** (hypercycle threshold)**.** *(a)* ***[Certified]*** *(Appendix E and* `code/single_root_cert.py`, `code/region_C_cert.py`*) For every $\kappa \in (\kappa_c, 1)$, $\mathcal{F}(\pi)$ has exactly one zero $\ell_{u,\mathrm{hyp}}(\kappa)$ in $\ell_u \in (0,\infty)$, and $\partial_{\ell_u}\mathcal{F}(\pi) > 0$ there. Hence the row is spectrally stable iff $\ell \ge \ell_{\mathrm{hyp}}(\kappa) := \ell_{u,\mathrm{hyp}}(\kappa)/\sqrt{1-\kappa^2}$.*

*(b)* ***[Proved, given (a)]****; monotonicity* ***[Lean]****. $\ell_{\mathrm{hyp}}$ is real-analytic on $(\kappa_c, 1)$, $\ell_{u,\mathrm{hyp}}$ is strictly decreasing (that $\ell_{\mathrm{hyp}} = \ell_{u,\mathrm{hyp}}/\sqrt{1-\kappa^2}$ is decreasing is only* ***[Numerical]****: Table 2, Figure 1), $\ell_{u,\mathrm{hyp}}(\kappa) < \operatorname{arcosh}(1/(2\kappa^2-1))$, and as $\kappa \downarrow \kappa_c$, $\ell_{u,\mathrm{hyp}}(\kappa) = -\log(2\kappa^2-1) + o(1)$; in particular $\ell_{\mathrm{hyp}} \to \infty$ and $\ell_A - \ell_{\mathrm{hyp}} \to \sqrt{2}\log 2$.*

*(c)* ***[Lean]****, given (a) for the last step. At fixed arc spacing $\ell$, $\mathcal{F}(\pi)|_{\ell_u = \ell\sin\alpha} \to \frac{\tau^2}{2}\Phi(\tau)$, $\tau = 2\pi/\ell$, as $\alpha \to 0^+$; hence for every $\varepsilon > 0$ and all small $\alpha > 0$ the row is unstable at arc spacing $\ell_* - \varepsilon$ and stable at $\ell_* + \varepsilon$, and $\ell_{\mathrm{hyp}}(\kappa) \to \ell_* = 2\pi/\tau_\infty$ as $\kappa \to 1^-$.*

*Lean: (b)* `Fpi_div_strictMono`, `stable_mono` *(at fixed strip spacing, stability is monotone in $\kappa^2$); (c)* `HyperbolicRings.Horocycle.tendsto_F_pi`, `hasSum_gL`, `hyper_threshold_tendsto`.

*Proof.* (a) It suffices to show $\partial_{\ell_u}\mathcal{F}(\pi) > 0$ at every zero with $\alpha < \pi/4$: then $\mathcal{F}(\pi)(\alpha,\cdot)$ can only cross zero upwards, so it has at most one zero, and it has one because it is negative at small spacing (Lemma E.4: **[Lean]**; also Lemma E.2) and positive beyond $\ell_A$ (Theorem 6.4(c)). Zeros with $\alpha \geq \pi/4$ do not exist (Theorem 6.4(b)). The zeros are covered by three regions. For $\ell_u \geq 6$ and $0.25 \leq \ell_u \leq 6$ we use `code/single_root_cert.py` (python-flint/Arb): for $\ell_u \geq 6$, $\mathcal{F}/\sin^2\alpha = P_1 r + P_2 r^2 + E$ with $r = e^{-\ell_u}$, $P_1 = 2\cos 2\alpha$, explicit tail bounds $|E| + |rE_r| \leq 0.32 r^2$, and $-P_2 \in [1.95, 2.01]$ on the relevant $\alpha$-range, which forces $\partial_{\ell_u}\mathcal{F} > 0$ at a zero; for $0.25 \leq \ell_u \leq 6$, 15 225 adaptive boxes on each of which the Lambert series (7), with rigorous tail bounds, either excludes 0 or has $\partial_{\ell_u}\mathcal{F} > 0$. For $\ell_u < 0.25$ we use the dual series (Theorem 6.7) and the estimates of Appendix E, checked by `code/region_C_cert.py` (113 boxes).

(b) $\mathcal{F}(\pi)$ is real-analytic in $(\alpha, \ell_u)$ (a convergent Lambert series), so the transversal zero is analytic by the implicit function theorem. Every term of $\mathcal{F}(\pi)/\sin^2\alpha$ is strictly increasing in $K$ at fixed $c$ ($\partial_K \mathfrak{g}_\mathrm{o} = 2(c+1)/D^2$, $\partial_K \mathfrak{g}_\mathrm{e} = 2(2c^2 + c - 1 - 2K)/D^3$), so with (a) the zero moves down as $\kappa$ increases (if $\kappa_1 < \kappa_2$, then $\mathcal{F}(\pi)$ at $(\kappa_2, \ell_{u,\mathrm{hyp}}(\kappa_1))$ is positive, so $\ell_{u,\mathrm{hyp}}(\kappa_2) < \ell_{u,\mathrm{hyp}}(\kappa_1)$). The bound is Theorem 6.4(c), with strict inequality because $\mathcal{F}(\pi) > 0$ at $\ell_u = \operatorname{arcosh}(1/(2K-1))$. As $\kappa \downarrow \kappa_c$ the increasing limit $L = \lim \ell_{u,\mathrm{hyp}}(\kappa)$ is infinite: if it were finite, continuity would give $\mathcal{F}(\pi) = 0$ at $(\kappa_c, L)$, contradicting Theorem 6.4(b). Finally $\frac{2\mathcal{F}(\pi)}{4\sigma} = (2K-1)\mathfrak{q} + (8K^2-3)\mathfrak{q}^2 + O(\mathfrak{q}^3)$, $\mathfrak{q} = e^{-\ell_u}$, uniformly in $\alpha$ (expand $\mathfrak{q}^n/(1 \mp \mathfrak{q}^n)$ in (7); the Lambert coefficients of $\mathcal{F}(\pi)/\sigma$ are bounded by $8n^2$), and $8K^2 - 3 \to -1$, so at the zero $\mathfrak{q}_* = (2K-1)(1 + o(1))$; the statement about $\ell_A$ follows from $\operatorname{arcosh} y = \log 2y + o(1)$.

(c) Write $\sigma = \sin^2\alpha$, $d = j\ell$, $c_j = \cosh(d\sin\alpha)$ and $w_j = (c_j - 1)/\sigma \geq d^2/2$. By Lemma 6.2 the $j$-th term of $\mathcal{F}(\pi)$ is $\frac{w-2c}{w(w+2)}$ ($j$ odd) or $\frac{3w-4c+6-4\sigma}{(w+2)^2}$ ($j$ even); both are bounded by $14/d^2$ and tend, as $\alpha \to 0^+$ ($w_j \to d^2/2$, $c_j \to 1$), to $-\frac{2}{d^2} + \frac{4}{d^2+4}$ and $\frac{2(3d^2+4)}{(d^2+4)^2}$. By dominated convergence (Tannery's theorem) $\mathcal{F}(\pi)$ tends to the sum $L(\ell)$ of these limits. Termwise, $2L(\ell)$ is the $yy$-entry of the horocycle-row Bloch matrix at the pairing mode $\theta = \frac{1}{2}$ (pair terms $-\frac{2}{d^2}$ for odd $j$ and $\frac{2(d^2-4)}{(d^2+4)^2}$ for even $j$, plus the momentum term $\frac{4}{d^2+4}$), so $L(\ell) = \frac{\tau^2}{2}\Phi(\tau)$ by Theorem 5.4(a). Its sign is that of $\Phi(\tau)$, negative for $\ell < \ell_*$ and positive for $\ell > \ell_*$ (Lemma 5.1); with (b) and (a) the unique zero converges. □

We write

$$I(\ell) := \frac{\tau^2}{2}\,\Phi(\tau) = \frac{\tau}{2}\coth\tau - \frac{\tau^2}{16}\coth^2\frac{\tau}{2} - \frac{1}{4}, \qquad \tau = \frac{2\pi}{\ell} \tag{8}$$

(the second form from (4) and $\tanh x + \coth x = 2\coth 2x$). It is also the $\alpha \to 0$ limit of the Lambert series (7) read as a Riemann sum ($n\alpha \to t$), i.e. the integral $\int_0^\infty [(2\sin 2t - 2t\sin^2 t)/(e^{\ell t} - 1) - 2t\cos^2 t/(e^{\ell t} + 1)]\,dt$; that form agrees with (8) to 15 digits by quadrature (**[Numerical]**, `code/rows_fpi.py`) and is not used. The limit says that the hypercycle family and the ring family share the same limit function $\Phi$: *the hypercycle rows, the horocycle row and the large rings all lose stability through the same pairing stiffness.*

**Theorem 6.7** (dual series; **[Proved]**, Appendix D)**.** *For $0 < \alpha < \pi$ and $\ell_u > 0$, with $\tilde{\mathfrak{q}} = e^{-2\pi^2/\ell_u}$,*

$$\mathcal{F}(\pi) = \mathcal{F}^{\mathrm{mod}} - \frac{2\pi}{\ell_u}\sum_{n\geq 1}\frac{\tilde{\mathfrak{q}}^{2n}}{1-\tilde{\mathfrak{q}}^{2n}}\Big[\sin 2\alpha \sinh\frac{4\pi n\alpha}{\ell_u} - \frac{4\pi n\sin^2\alpha}{\ell_u}\Big]$$

| $\kappa$ | 0.72 | 0.75 | 0.80 | 0.90 |
|---|---|---|---|---|
| $\ell_{\mathrm{hyp}}(\kappa)$ | 4.7635102682 | 3.1863292497 | 2.2554766253 | 1.4365846813 |
| $\kappa$ | 0.95 | 0.99 | 0.999 | 0.9999 |
| $\ell_{\mathrm{hyp}}(\kappa)$ | 1.1694721423 | 0.9537651073 | 0.8740556005 | 0.8530342205 |

Table 2: Hypercycle thresholds (arc spacing), the zero of $\mathcal{F}(\pi)$, **[Numerical]** (`code/rows_thresholds.py`); the one-variable series is evaluated at 30 digits, but the root is not enclosed in an interval. Near $\kappa_c$, $\ell_{\mathrm{hyp}} \approx -\log(2\kappa^2-1)/\sqrt{1-\kappa^2}$ (relative error $3\cdot 10^{-6}$ at $\kappa = 0.708$).

$$- \frac{2\pi^2 \sin^2\alpha}{\ell_u^2} \sum_{n\ge 1} \frac{n\tilde{\mathfrak{q}}^n}{1-\tilde{\mathfrak{q}}^n}\Big[1+\cosh\frac{2\pi n\alpha}{\ell_u}\Big],$$

$$\mathcal{F}^{\mathrm{mod}} = \frac{\pi \sin 2\alpha}{2\ell_u}\coth\frac{2\pi\alpha}{\ell_u} - \frac{\pi^2\sin^2\alpha}{4\ell_u^2}\coth^2\frac{\pi\alpha}{\ell_u} - \frac{\alpha\sin 2\alpha}{\ell_u} - \frac{\cos 2\alpha}{4}$$

*For $0<\alpha<\alpha_0$ ($\tan\alpha_0 = 2\alpha_0$, so for all $\kappa > \cos\alpha_0 \approx 0.394$) every correction term is negative, so $\mathcal{F}(\pi) < \mathcal{F}^{\mathrm{mod}}$.*

The sign statement follows from $\sinh y \ge y$: each bracket of the first sum is $\ge \frac{4\pi n\sin\alpha}{\ell_u}(2\alpha\cos\alpha - \sin\alpha) > 0$. At fixed arc spacing the corrections are $O(\alpha^{-1}e^{-2\pi^2/(\ell\sin\alpha)})$, and at $\ell_u = \ell\sin\alpha$, $\mathcal{F}^{\mathrm{mod}} \to I(\ell)$ as $\alpha\to 0$ (put $\alpha = 0$ in each term). Four independent evaluations (lattice sum, Lambert series, theta form, dual series) agree to $10^{-45}$ at ten points (`code/rows_fpi.py`), and each step of the proof is checked numerically in `code/dual_series_check.py`. The elementary equation $\mathcal{F}^{\mathrm{mod}} = 0$ reproduces $\ell_{\mathrm{hyp}}(\kappa)$ to $10^{-4}$ at $\kappa = 0.8$ and $10^{-11}$ at 0.9 **[Numerical]**.

## 7 At the horocycle point

Figure 1 suggests one continuous boundary through $(\kappa,\ell) = (1,\ell_*)$, with circles ($\kappa = \coth r > 1$) on the right and hypercycles on the left. The rings and the hypercycle rows do meet there, but the boundary is not one analytic curve.

**Theorem 7.1** (three germs at the horocycle point)**.** *Let $\kappa_N = \coth r_*(N)$ and $\ell_N = 2\pi\sinh r_*(N)/N$ be the threshold ring of $N$ vortices.*

*(a)* ***[Proved; rates in Lean]*** *$(\kappa_N,\ell_N)\to(1,\ell_*)$, with $\kappa_N - 1 = \tau_\infty^2/(2N^2) + O(N^{-4})$ and $\ell_N - \ell_* = O(N^{-2})$; $(1,\ell_*)$ is the only accumulation point of the ring thresholds in $[1,\infty)\times(0,\infty)$, and it is the endpoint of the hypercycle branch (Theorem 6.6(c)).*

*(b)* ***[Proved, given Theorem 6.6(a)]*** *Write $\kappa = \cos\alpha$. The zero $\ell_{\mathrm{mod}}(\alpha)$ of $\mathcal{F}^{\mathrm{mod}}(\ell\sin\alpha,\alpha)$ near $\ell_*$ is real-analytic in $\alpha$ at 0 with $\ell_{\mathrm{mod}} = \ell_* + \lambda_1\alpha + O(\alpha^2)$, $\lambda_1 = 2/(\ell_* I'(\ell_*)) > 0$, and $0 < \ell_{\mathrm{hyp}}(\cos\alpha) - \ell_{\mathrm{mod}}(\alpha) = O\big(\alpha^{-1}e^{-(2\pi^2/\ell_* - \eta)/\alpha}\big)$ for every $\eta > 0$. Hence $\ell_{\mathrm{hyp}}(\kappa) = \ell_* + \lambda_1\sqrt{2(1-\kappa)} + O(1-\kappa)$: the hypercycle branch has a square-root branch point at $\kappa = 1$. Moreover $\alpha\mapsto\ell_{\mathrm{hyp}}(\cos\alpha)$ is not real-analytic at $\alpha = 0$: no function analytic on a neighbourhood of 0 agrees with it on an interval $(0,\delta)$ (nor does any $\Lambda\circ v$ with $\Lambda$, $v$ analytic at 0).*

*(c)* ***[Proved; the sign in Lean; decimals Numerical]*** *The ring thresholds approach with finite slopes depending on the parity:*

$$\frac{\ell_N - \ell_*}{\kappa_N - 1} \to \mathfrak{m}_{\mathrm{e,o}} = \frac{4\pi(\varsigma_{\mathrm{e,o}} - \tau_\infty/2)}{\tau_\infty^2}, \qquad \mathfrak{m}_{\mathrm{o}} - \mathfrak{m}_{\mathrm{e}} = -\frac{4\pi\, p(\tau_\infty)}{|\Phi'(\tau_\infty)|\,\tau_\infty^4} < 0,$$

*numerically* $\mathfrak{m}_{\mathrm{e}} = -0.72122$, $\mathfrak{m}_{\mathrm{o}} = -0.75300$, $\lambda_1 = 0.632452$.

*Consequently no curve that is differentiable in $\kappa$ at $\kappa = 1$ contains both the hypercycle branch and the threshold rings, and no curve with a finite slope at $\kappa = 1$ contains the threshold rings of both parities.*

*Proof.* (a) $\coth r_* = \cosh(\tau_*/N)$ because $\cosh r = \coth(\tau/N)$; with $\tau_* = \tau_\infty + O(N^{-2})$ (Theorem 5.2) this gives the rate. A ring threshold point $(\kappa_N, \ell_N)$ with $N$ bounded is isolated from $\kappa = 1$, and $\kappa_N > 1$.

(b) $\mathcal{F}^{\mathrm{mod}}(\ell \sin\alpha, \alpha)$ is analytic in $(\ell, \alpha)$ near $(\ell_*, 0)$ (the arguments of coth stay away from 0), equals $I(\ell)$ at $\alpha = 0$, and its only odd part in $\alpha$ is $-\alpha \sin 2\alpha/\ell_u = -2\alpha\cos\alpha/\ell$, so $\partial_\alpha \mathcal{F}^{\mathrm{mod}} = -2/\ell$ at $\alpha = 0$. By (8), $I'(\ell_*) = -\tau_\infty^3 \Phi'(\tau_\infty)/(2\ell_*) > 0$ (Lemma 5.1), and the implicit function theorem gives $\ell_{\mathrm{mod}}$ with slope $\lambda_1 = 2/(\ell_* I'(\ell_*))$. Since every correction term of Theorem 6.7 is negative for small $\alpha$, $\mathcal{F}(\pi) < \mathcal{F}^{\mathrm{mod}}$, so at $\ell_{\mathrm{mod}}$ the function $\mathcal{F}(\pi)$ is negative and its unique zero (Theorem 6.6(a)) lies above; the size follows from the correction bound and transversality. If $\ell_{\mathrm{hyp}}(\cos\alpha)$ agreed on $(0, \delta)$ with a function analytic at 0 (in particular with $\Lambda(v(\alpha))$), then $\ell_{\mathrm{hyp}} - \ell_{\mathrm{mod}}$ would be analytic at 0 with all derivatives 0 there (it is $O(e^{-c/\alpha})$), hence identically 0 near 0, contradicting strict positivity. Finally $\alpha = \arccos\kappa = \sqrt{2(1-\kappa)}(1 + O(1-\kappa))$.

(c) By Proposition 5.3, $\ell_N - \ell_* = 2\pi(\varsigma - \tau_\infty/2)N^{-2} + O(N^{-4})$, and $\kappa_N - 1 = \tau_\infty^2/(2N^2) + O(N^{-4})$. The difference of the slopes is $4\pi(\varsigma_{\mathrm{o}} - \varsigma_{\mathrm{e}})/\tau_\infty^2$ with $\varsigma_{\mathrm{o}} - \varsigma_{\mathrm{e}} = -p(\tau_\infty)/(|\Phi'(\tau_\infty)|\tau_\infty^2)$, and $p(\tau_\infty) > 0$, $\Phi'(\tau_\infty) < 0$ are in Lean (`psiXX_tauInf_neg`, `Phi_strictAntiOn`).

The consequence: along the hypercycle branch $(\ell - \ell_*)/(1-\kappa) \to +\infty$, while along each ring sequence it converges. □

*Remark* 7.2 (ring curves; [**Numerical**] here). Numerically, near $(1, \ell_*)$ the even and the odd ring thresholds lie on two real-analytic curves $\ell^{\mathrm{e}}(\kappa)$ and $\ell^{\mathrm{o}}(\kappa)$, the zero sets of the even part of $\mathcal{F}^{\mathrm{mod}}$ continued to imaginary $\alpha$ (with $\hat{m}^2 = -\ell^2 \sin^2\alpha/4\pi^2$ in the odd case): the rings $N = 8, 9, 16, 17, 64, 65$ lie on their parity's curve to $10^{-51}$ and miss the other by $10^{-2}$–$10^{-4}$ (`code/b7_curves.py`). The continued hypercycle germ $\ell_{\mathrm{mod}}(iy)$ is not real (its imaginary part is $\approx \lambda_1 y$), so it misses the rings at order $N^{-1}$.

## 8 Relation to prior work

**Boatto [2].** Boatto posed exactly this problem for this model. She uses the true hyperbolic Green function (her eq. (14), $2\pi\Psi = -\ln\tanh(\sqrt{|K_{\mathrm{B}}|}\, r/2)$; eq. (17)), fixes the geodesic radius $r_0 = 1$ and varies the curvature $K_{\mathrm{B}}$, writes the Hessian of the augmented Hamiltonian in circulant block form with a coupling block (her Lemma 5.2, eqs. (41)–(45)), and uses the Dirichlet (definiteness) criterion. She evaluates the extreme eigenvalues numerically: her Fig. 5a plots the minimum eigenvalue for $N = 5, \ldots, 9$ over about $-0.6 < K_{\mathrm{B}} < 0.6$ (Fig. 5b, over $-0.75 < K_{\mathrm{B}} < 0.75$, is $N = 6$ only). Fig. 5a already shows the curves for $N \le 7$ positive for all plotted $K_{\mathrm{B}} < 0$, and the zero crossings for $N = 8$ and $N = 9$, read off the plotted curves near $K_{\mathrm{B}} \approx -0.25$ and $-0.4$; evaluating her equations precisely gives $-0.2822$ and $-0.489$ (Remark 3.8), i.e. our $\xi_*(8)$ and $\xi_{\mathrm{EM}}(9)$. Her text reads "In particular for $N \ge 7$, we can only assure stability for negative $K$" in the preprint (mp_arc 06-311, 2006) and "... for sufficiently negative $K$" in the journal version, whose conclusion adds that "a ring of $N \ge 7$ can only be assured to be nonlinearly stable for sufficiently negative values of $K$". We read both versions; the figures are the same, and equation numbers above are those of the journal version. Credit for the question, the block structure and the qualitative picture is hers. Her equations reproduce our thresholds exactly (Remark 3.8). New here: closed-form blocks and the exact symbol; proofs that $N \le 7$ is stable at every radius and that each $N \ge 8$ has exactly one

threshold; certified values; the spectral (rather than definiteness) analysis and the odd-$N$ Krein window that a definiteness criterion cannot see; the large-$N$ asymptotics; and everything about rows.

**Montaldi–Tokieda [11].** Their §2 (eq. (2.11), Theorem 2) gives a closed-form criterion for rings on surfaces of constant curvature with the log-chordal kernel *with* a uniform background (the sinh model on $\mathbb{H}^2$) and proves "$n \leq 7$ always stable; $n > 7$ stable for $a$ sufficiently large"; their §3 studies the bifurcations. In §4.2 they consider the image Green function (Kimura's Hamiltonian, the present model) and write that the "calculations ... get so cumbersome that the stability problem seems no longer tractable analytically". Theorem 3.6 is the analogue of their Theorem 2 for this model; the exact symbol (Theorem 3.3) is what makes it tractable. The large-$N$ behaviour differs: their thresholds have $\cosh r_* \sim \sqrt{N}/2$, ours $\cosh r_* \sim N/\tau_\infty$.

**Other work on $\mathbb{H}^2$.** Kimura [7] introduced the model. Montaldi and Nava-Gaxiola [10] (and [12]) treat $N = 2, 3$ on $\mathbb{H}^2$; Hwang and Kim treat three vortices [4] and the existence of ring relative equilibria [5]. None of these addresses the stability thresholds of $N$-rings for general $N$.

**Rows.** The planar single row is unstable at every spacing [17, 8, 1]. In the upper half-plane our pair interaction is a row plus its mirror row, with the hyperbolic symplectic form, so the stability of the horocycle row above $\ell_*$ (Theorem 5.4) is a curvature effect. We have not found prior work on vortex rows along horocycles or hypercycles in $\mathbb{H}^2$, on the critical curvature $1/\sqrt{2}$, or on the relation between ring thresholds and row thresholds.

# 9 Status and limitations

- **What Lean checks.** The ring results of §§2–5 are checked in Lean from the Hamiltonian (1) (criticality, the Hessian as circulant matrices, its Fourier blocks, the spectrum of the full $2N$-dimensional linearisation as the union of block spectra) through the thresholds, the Krein window and the large-$N$ rates; the horocycle row is checked from its pair Hessian and four lattice sums. For hypercycle rows Lean starts from the pair term (5) (proved equal to the upper-half-plane tanh pair term) and derives the Bloch symbol, the spectral criterion, the critical mode $q = \pi$, the critical curvature (Theorem 6.4), monotonicity in the curvature, and the convergence of the hypercycle rows to the horocycle row (Theorem 6.6(c), up to the certified uniqueness). Lean does *not* check that the linear operators it analyses are the derivative of the vortex vector field (for rings it is defined as $\omega^{-1} d^2 F$; for rows, as the lattice sum of per-pair second derivatives, with Hamilton's equations in the strip chart taken as given), nor any nonlinear statement. The only axioms used are `propext`, `Classical.choice` and `Quot.sound` (`outputs/lean_axioms.out`, 120 theorems). The full library builds in about 21 minutes on 12 cores; the certificate modules for $N = 14, 15, 16$ take 4–12 minutes each.

- **Not formalised.** The single hypercycle threshold (Theorem 6.6(a)) rests on Arb certificates with about 15 000 boxes (regions with $\ell_u \geq 0.25$) and 113 boxes (region $\ell_u < 0.25$, Appendix E); formalising them would need verified interval arithmetic for Lambert series in Lean, which we have not attempted. The dual series (Theorem 6.7, Appendix D) is proved by hand: it needs the Fourier transform of $1/(\cosh x - \cos\theta)$ by contour integration and the Poisson summation formula; Mathlib has Poisson summation but not the residue computation of this transform, so we did not formalise it. Analyticity of $\ell_{\mathrm{hyp}}$ and the law near

$\kappa_c$ (Theorem 6.6(b)), Proposition 5.3 and Theorem 7.1(b) use the implicit function theorem for real-analytic functions (Mathlib has a $C^1$ implicit function theorem but not the analytic one in the form needed). Lemma E.4 (dense rows are unstable) is in Lean; the estimates of Lemmas E.1–E.3 are by hand and checked in the certificate.

- **Numerical items.** Table 2, the decrease of $\ell_{\mathrm{hyp}}$ in $\kappa$, the decimals of $\varsigma_{\mathrm{e,o}}$, $\mathfrak{m}_{\mathrm{e,o}}$, $\lambda_1$, Remark 7.2 and the Krein-window constant are floating point (30–50 digits, no interval enclosure).
- **Nonlinear stability** is not addressed. In the odd-$N$ window it is genuinely open; at the thresholds themselves the linearisation is degenerate.
- **Scope.** Only equal circulations and the kernel $-\log\tanh(d/2)$; screened kernels are mentioned in Remark 6.5. The degree pattern of Remark 3.7 is established for $N \le 16$ only.
- **Review.** The results were developed with extensive computer checking and an AI-assisted adversarial review, but no human specialist in point-vortex dynamics has refereed them.


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## A Proof of the block theorem

Since $dF = 0$ at the ring (Lemma 2.2), the Hessian is coordinate-independent. We add the three terms of (2) and $-\Omega J$.

*Pair term.* As for the planar ring, $-\sum_{j<k}\log|z_j - z_k|$ contributes $-\frac{m(N-m)}{2}$ to the $ss$ entry and $+\frac{m(N-m)}{2}$ to the $\theta\theta$ entry on mode $m$, with no coupling (it depends on $z_j - z_k = \sqrt{\xi}(w^j e^{\epsilon_j} - w^k e^{\epsilon_k})$ only through log of a holomorphic function of $\epsilon = s + i\theta$, whose real part has trace-free Hessian).

*One-body terms.* $V(s) = -\frac{1}{2}\log(1-u) - \Omega\frac{1+u}{1-u}$, $u = \xi e^{2s}$, is independent of $\theta$; $d/ds = 2u\,d/du$ gives $V''(0) = \frac{2\xi}{(1-\xi)^2} - 2c\,\Omega\rho_0$, which with Lemma 2.2 and $c - 1 = 2\xi/(1-\xi)$ equals $(N-1)c - \frac{1}{2}(c-1)^2 + \frac{2NcX}{1-X}$; it enters only the $ss$ entry.

*Power sums.* With $|P_p|^2 = \xi^p \sum_{j,k} w^{(j-k)p} e^{p\epsilon_j + p\bar{\epsilon}_k}$, the quadratic part of $\Pi = -\frac{1}{2}\sum_p |P_p|^2/p$ is $\Pi_2 = -\frac{N}{2}W_N \sum_j (s_j^2 - \theta_j^2) - \frac{1}{2}\sum_p p\,\xi^p \big|\sum_j w^{jp}(s_j + i\theta_j)\big|^2$. The first piece gives $\mp N W_N$ on the diagonal. In the second, $\sum_k w^{(j-k)p} w^{mk} = N w^{mj}[p \equiv m]$ gives the symbols $N([p \equiv m] + [p \equiv -m])$ on both diagonal entries and $-iN([p \equiv m] - [p \equiv -m])$ off the diagonal; summing $-\frac{1}{2}p\xi^p$ times these over $p$ gives $-\frac{N}{2}(W_m + W_{N-m})$ and $ig_m$. Adding the pieces gives the theorem. □

## B Proof of the critical-mode lemma

Write $h = x\tau \in (0, \tau]$, $T = \tau\coth\tau$, $S(t) = \sinh t/t$. A direct computation gives $4\sinh^2\tau\,\partial_x\psi = \sinh\tau\, v(h)$ with $v(h) = (T-1)\sinh h - h\cosh h - hS(\tau)$, so we need $T - 1 - h\coth h < S(\tau)/S(h)$. Let $\lambda = \log S$; then $\lambda' = \coth t - 1/t$, $\lambda'' = t^{-2} - \sinh^{-2} t \in [0, \min(\frac{1}{3}, t^{-2})]$ and $(t\coth t)' = \lambda' + t\lambda''$.

With $D=\lambda(\tau)-\lambda(h)\ge 0$ and $I=\int_h^\tau t\lambda''\,dt$, $T-h\coth h=D+I$ and the claim is $I<1+e^D-D$. The right side is increasing in $D$ and exceeds 2 for $h<\tau$, while $I\le\int_0^\tau t\min(\frac13,t^{-2})dt\le 2$ for $\tau\le\sqrt{3}e^{3/2}$. For larger $\tau$: if $h\ge\tau/2$ then $I\le\log 2$; if $h<\tau/2$ then $e^D\ge\cosh(\tau/2)$ and $1+\frac12 e^{\tau/2}-\frac{\tau}{2}>\frac12+\log(\tau/\sqrt{3})\ge I$. □

# C Derivation of the row symbol

Write $H=\sum_{i<j}W(X_{ij})$ and linearise $\dot u_0=\sin^2\varphi_0\,\partial_{\varphi_0}H$, $\dot\varphi_0=-\sin^2\varphi_0\,\partial_{u_0}H$ at the row, with $u_k=k\ell_u+U_k$, $\varphi_k=\alpha+P_k$. Each pair $(0,j)$ contributes a $2\times4$ block depending on $u_0-u_j=\mp j\ell_u$ and on $(\varphi_0,\varphi_j)$. By the reflection $u\mapsto-u$, contributions to $\dot u_0$ are even in $j$ under $U$-variations and odd under $P$-variations, and conversely for $\dot\varphi_0$. Summing over $j\neq0$ with Bloch phases $e^{iqj}$, the even parts give the off-diagonal entries and the odd parts give the equal diagonal entries $i\varpi(q)$. Explicitly, with the pair functions evaluated at $X=X_{0j}$,

$$B^{(0)}=\partial_{\varphi_0}\big(\sin^2\varphi_0\,\partial_{\varphi_0}W\big),\qquad B^{(1)}=\sin^2\alpha\,\partial_{\varphi_0}\partial_{\varphi_j}W,\qquad C^{(1)}=-\sin^2\alpha\,\partial_{u_0}^2W,$$

the $\delta\varphi$-column of $\dot u_0$ is $\sum_{j\neq0}[B^{(0)}+B^{(1)}e^{iqj}]$ (the drift term $\partial_{\varphi_0}\sin^2\varphi_0\cdot\partial_{\varphi_0}H$ is part of $B^{(0)}$), and the $\delta u$-column of $\dot\varphi_0$ is $\sum_{j\neq0}C^{(1)}(1-e^{iqj})$ (translation invariance gives the $1-e^{iqj}$). Differentiating $X_{0j}$ gives the generic forms quoted after the proposition, and $W=\frac12\log\frac{X+1}{X-1}$ the rational ones. Since $C^{(1)}<0$, $C(q)<0$ for $q\neq0$, and $(\lambda-i\varpi)^2=BC$ gives the exponents.

In terms of the pair term (5) at the row pair $(0,\alpha,b,\alpha)$, $b=k\ell_u$, with $D_+=\cosh b-\cos2\alpha$ and $D_-=\cosh b-1$, the eight coefficients are

$$\mathrm{on}_B=\sin\alpha\cos\alpha\,\frac{\sin2\alpha}{D_+}+\frac{\sigma}{2}\Big[\frac{\cos2\alpha\,D_+-\sin^2 2\alpha}{D_+^2}-\frac{1}{D_-}\Big],$$

$$\mathrm{cp}_B=\frac{\sigma}{2}\Big[\frac{\cos2\alpha\,D_+-\sin^2 2\alpha}{D_+^2}+\frac{1}{D_-}\Big],\qquad \mathrm{on}_U=-\mathrm{cp}_U=-\mathrm{cp}_P=\frac{\sigma\sin2\alpha\sinh b}{2D_+^2},$$

$$\mathrm{on}_C=-\mathrm{cp}_C=-\frac{\sigma}{2}\Big[\frac{\cosh b\,D_+-\sinh^2 b}{D_+^2}-\frac{\cosh b\,D_--\sinh^2 b}{D_-^2}\Big],$$

$$\mathrm{on}_P=\sin\alpha\cos\alpha\sinh b\Big(\frac{1}{D_+}-\frac{1}{D_-}\Big)-\mathrm{on}_U,$$

with $\mathrm{on}_B=B^{(0)}$, $\mathrm{cp}_B=B^{(1)}$, $\mathrm{on}_C=C^{(1)}$ at $K=\cos^2\alpha$, $c=\cosh b$. The odd ones cancel in pairs $k\leftrightarrow-k$ except for $\varpi(q)=-2\sum_{j\ge1}\mathrm{on}_U(j\ell_u)\sin qj$, which appears on both diagonal entries. These are the Lean theorems `onB_val`, …, `cpP_val`, `onB_eq_B0`, `cpB_eq_B1`, `onC_eq_C1`. The computation is also carried out in sympy in `code/rows_symbol.py`, part (a), and compared there with a numerical Hessian of the pair energy (part (c)), the lattice-sum symbol (part (d)) and the eigenvalues of a 6-site periodic supercell (part (e)).

# D Proof of the dual series (Theorem 6.7)

Throughout, $\ell:=\ell_u>0$, $\sigma=\sin^2\alpha$, $C_k=\cosh(k\ell)$, $\mathfrak{q}=e^{-\ell}$.

**Step 1: from Lambert series to lattice sums.** Using $\frac{x^n}{1+x^n}=\frac{x^n}{1-x^n}-\frac{2x^{2n}}{1-x^{2n}}$ and $\sin^2+\cos^2=1$ in (7),

$$\mathcal{F}(\pi)=\sin2\alpha\sum_{n\ge1}\frac{\sin(2n\alpha)\mathfrak{q}^n}{1-\mathfrak{q}^n}-2\sigma\sum_{n\ge1}\Big[\frac{n\mathfrak{q}^n}{1-\mathfrak{q}^n}-\frac{n\mathfrak{q}^{2n}}{1-\mathfrak{q}^{2n}}\Big]+2\sigma\sum_{n\ge1}\frac{n\cos(2n\alpha)\mathfrak{q}^{2n}}{1-\mathfrak{q}^{2n}}.$$

Expand $\frac{\mathfrak{q}^n}{1-\mathfrak{q}^n}=\sum_{k\ge1}\mathfrak{q}^{nk}$ (all double series converge absolutely) and use, for $x=e^{-t}$, $t>0$,

$$\sum_{n\ge1}\sin(n\theta)x^n=\frac{\sin\theta}{2(\cosh t-\cos\theta)},\qquad \sum_{n\ge1}nx^n=\frac{1}{2(\cosh t-1)},$$
$$\sum_{n\ge1}n\cos(n\theta)x^n=\frac{\cos\theta\cosh t-1}{2(\cosh t-\cos\theta)^2},$$

the last being the $\theta$-derivative of the first. With $g_\theta(x):=1/(\cosh x-\cos\theta)$ and $S_\theta(\ell):=\sum_{k\ge1}g_\theta(k\ell)$ (continuous in $\theta\in[0,2\pi)$ for $\ell>0$, and $C^1$ in $\theta$ by uniform convergence of the derivative), and noting $\frac{\cos\theta C-1}{(C-\cos\theta)^2}=\cos\theta\,g_\theta(x)+\sin\theta\,\partial_\theta g_\theta(x)$ at $C=\cosh x$, we get

$$\mathcal{F}(\pi)=\frac{\sin^2 2\alpha}{2}S_{2\alpha}(\ell)-\sigma\big[S_0(\ell)-S_0(2\ell)\big]+\sigma\big[\cos2\alpha\,S_{2\alpha}(2\ell)+\sin2\alpha\,\partial_\theta S_\theta(2\ell)|_{\theta=2\alpha}\big].\qquad(9)$$

**Lemma D.1** (Fourier transform)**.** *For $0<\theta<2\pi$ and real $\omega$, $\displaystyle\int_{\mathbb{R}}\frac{e^{-i\omega x}}{\cosh x-\cos\theta}\,dx=\frac{2\pi\sinh((\pi-\theta)\omega)}{\sin\theta\,\sinh(\pi\omega)}$ (for $\omega=0$: $2(\pi-\theta)/\sin\theta$).*

*Proof.* Integrate $f(z)=e^{-i\omega z}/(\cosh z-\cos\theta)$ around the rectangle with vertices $\pm X$, $\pm X+2\pi i$. The vertical sides contribute $O(e^{-X})$, and on the top side $f(x+2\pi i)=e^{2\pi\omega}f(x)$. The poles in the strip are $z=i\theta$ and $z=i(2\pi-\theta)$, with residues $e^{\omega\theta}/(i\sin\theta)$ and $-e^{\omega(2\pi-\theta)}/(i\sin\theta)$. Hence $(1-e^{2\pi\omega})\int_{\mathbb{R}}f=2\pi(e^{\omega\theta}-e^{\omega(2\pi-\theta)})/\sin\theta$, which is the claim. □

**Lemma D.2** (dual form of $S_\theta$)**.** *For $0<\theta<2\pi$ and $\ell>0$, with $Q=e^{-4\pi^2/\ell}$,*

$$S_\theta(\ell)=\frac{\pi\coth(\pi\theta/\ell)-\theta+E_\theta(\ell)}{\ell\sin\theta}-\frac{1}{2(1-\cos\theta)},\qquad E_\theta(\ell)=-4\pi\sum_{m\ge1}\frac{Q^m\sinh(2\pi\theta m/\ell)}{1-Q^m}.$$

*Moreover $S_0(\ell)-S_0(2\ell)=\frac{\pi^2}{4\ell^2}-\frac{1}{2\ell}+\frac{E_0'(\ell)}{\ell}-\frac{E_0'(2\ell)}{2\ell}$, where $E_0'=\partial_\theta E_\theta|_{\theta=0}$.*

*Proof.* $g_\theta$ is even, and it and all its derivatives are $O(e^{-|x|})$, so $g_\theta\in\mathcal{S}(\mathbb{R})$; and by Lemma D.1 its transform is $O(e^{-\min(\theta,2\pi-\theta)|\omega|})$, so the Poisson summation formula $\sum_{k\in\mathbb{Z}}g_\theta(k\ell)=\ell^{-1}\sum_{m\in\mathbb{Z}}\hat g_\theta(2\pi m/\ell)$ holds [15, Ch. 5, §3] (stated there with the kernel $e^{-2\pi ix\xi}$; the form above follows by rescaling $x\mapsto\ell x$ and $\xi=\omega/2\pi$). With $a=2\pi/\ell$, $x=e^{-\theta a}$, $y=e^{-(2\pi-\theta)a}$ and $xy=e^{-2\pi a}=Q$,

$$\sum_{m\ge1}\frac{\sinh((\pi-\theta)ma)}{\sinh(\pi ma)}=\sum_{m\ge1}\frac{x^m-y^m}{1-Q^m}=\frac{x}{1-x}+\sum_{m\ge1}\Big[\frac{x^mQ^m}{1-Q^m}-\frac{y^m}{1-Q^m}\Big],$$

and $2\pi\frac{x}{1-x}=\pi\coth\frac{\theta a}{2}-\pi$, while $x^mQ^m-y^m=Q^m(e^{-\theta am}-e^{\theta am})$. Collecting, $\sum_{\mathbb{Z}}g_\theta(k\ell)=\frac{2}{\ell\sin\theta}\big[\pi\coth\frac{\pi\theta}{\ell}-\theta+E_\theta(\ell)\big]$, and $S_\theta=\frac12[\sum_{\mathbb{Z}}-g_\theta(0)]$ with $g_\theta(0)=1/(1-\cos\theta)$. For the limit $\theta\to0$: $E_0=0$, $E_\theta=E_0'\theta+O(\theta^3)$ (odd in $\theta$), and $\pi\coth(\pi\theta/\ell)=\ell/\theta+\pi^2\theta/(3\ell)+O(\theta^3)$, so $\frac{\pi\coth(\pi\theta/\ell)-\theta+E_\theta(\ell)}{\ell\sin\theta}=\frac{1}{\theta^2}+\frac{\pi^2}{3\ell^2}-\frac{1}{\ell}+\frac16+\frac{E_0'(\ell)}{\ell}+O(\theta^2)$. Subtracting the same expression at $2\ell$ (the singular terms and $g_\theta(0)$ cancel) gives the formula. □

*Proof of Theorem 6.7.* Insert Lemma D.2 into (9). *Main parts.* The first term gives $\frac{\sin2\alpha}{2\ell}(\pi\coth\frac{2\pi\alpha}{\ell}-2\alpha)$ and $-\frac{\sin^2 2\alpha}{4(1-\cos2\alpha)}=-\frac{\cos^2\alpha}{2}$; the second gives $-\sigma(\frac{\pi^2}{4\ell^2}-\frac{1}{2\ell})$. In the

third, for any $C^1$ function $u$, $\cos\theta\,\frac{u(\theta)}{2\ell\sin\theta}+\sin\theta\,\partial_\theta\frac{u(\theta)}{2\ell\sin\theta}=\frac{u'(\theta)}{2\ell}$; with $u=\pi\coth\frac{\pi\theta}{2\ell}-\theta$ this gives $\sigma(-\frac{\pi^2}{4\ell^2}\operatorname{csch}^2\frac{\pi\alpha}{\ell}-\frac{1}{2\ell})$, and the $-\frac{1}{2(1-\cos\theta)}$ part gives $-\frac{\cos 2\alpha}{4}+\frac{\cos^2\alpha}{2}$. The sum is $\frac{\pi\sin 2\alpha}{2\ell}\coth\frac{2\pi\alpha}{\ell}-\frac{\alpha\sin 2\alpha}{\ell}-\frac{\pi^2\sigma}{4\ell^2}\big(1+\operatorname{csch}^2\frac{\pi\alpha}{\ell}\big)-\frac{\cos 2\alpha}{4}=\mathcal{F}^{\mathrm{mod}}$. *Corrections.* With $Q(\ell)=\tilde{\mathfrak{q}}^2$ and $Q(2\ell)=\tilde{\mathfrak{q}}$: the first term gives $\frac{\sin 2\alpha}{2\ell}E_{2\alpha}(\ell)=-\frac{2\pi\sin 2\alpha}{\ell}\sum_n\frac{\tilde{\mathfrak{q}}^{2n}\sinh(4\pi n\alpha/\ell)}{1-\tilde{\mathfrak{q}}^{2n}}$; the second, using $E_0'(\ell)=-\frac{8\pi^2}{\ell}\sum_n\frac{n\tilde{\mathfrak{q}}^{2n}}{1-\tilde{\mathfrak{q}}^{2n}}$ and $E_0'(2\ell)=-\frac{4\pi^2}{\ell}\sum_n\frac{n\tilde{\mathfrak{q}}^{n}}{1-\tilde{\mathfrak{q}}^{n}}$, gives $\frac{8\pi^2\sigma}{\ell^2}\sum_n\frac{n\tilde{\mathfrak{q}}^{2n}}{1-\tilde{\mathfrak{q}}^{2n}}-\frac{2\pi^2\sigma}{\ell^2}\sum_n\frac{n\tilde{\mathfrak{q}}^{n}}{1-\tilde{\mathfrak{q}}^{n}}$; the third gives, by the same identity with $u=E_\theta(2\ell)$, $\frac{\sigma}{2\ell}\partial_\theta E_\theta(2\ell)|_{2\alpha}=-\frac{2\pi^2\sigma}{\ell^2}\sum_n\frac{n\tilde{\mathfrak{q}}^n\cosh(2\pi n\alpha/\ell)}{1-\tilde{\mathfrak{q}}^n}$. These are exactly the two correction sums of Theorem 6.7. Termwise differentiation and rearrangement are justified by absolute and locally uniform convergence. □

Every step is checked numerically (30 digits) in `code/dual_series_check.py`: the Fourier transform against quadrature, $S_\theta$ and the $\theta\to 0$ limit against direct sums, (9) and the final identity against the lattice sum.

# E The estimates for small strip spacing

Here $0<\alpha\le\pi/4$, $\ell_u<0.25$ and $\eta:=\pi\alpha/\ell_u$. By Theorem 6.7, $\mathcal{F}(\pi)=\mathcal{F}^{\mathrm{mod}}-R$ with $R=R_1+R_2\ge 0$; in the variables $(\alpha,\eta)$, with $\lambda=2\pi\eta/\alpha=2\pi^2/\ell_u$ and $\operatorname{sinc}x=\sin x/x$,

$$\mathcal{F}^{\mathrm{mod}}=\eta\operatorname{sinc}(2\alpha)\coth 2\eta-\frac{\eta^2}{4}\operatorname{sinc}^2\alpha\coth^2\eta-\frac{\eta}{\pi}\sin 2\alpha-\frac{\cos 2\alpha}{4},$$

$$R_1=\frac{2\eta}{\alpha}\sum_{n\ge1}\frac{e^{-2n\lambda}}{1-e^{-2n\lambda}}\Big[\sin 2\alpha\sinh 4n\eta-\frac{4n\eta\sin^2\alpha}{\alpha}\Big],$$
$$R_2=\frac{2\eta^2\sin^2\alpha}{\alpha^2}\sum_{n\ge1}\frac{ne^{-n\lambda}}{1-e^{-n\lambda}}\big[1+\cosh 2n\eta\big],$$

and at fixed $\alpha$, $\partial_{\ell_u}\mathcal{F}=-\frac{\eta}{\ell_u}\partial_\eta\mathcal{F}$, so a zero is transversal (upward in $\ell_u$) iff $\partial_\eta\mathcal{F}<0$ there. At fixed $\alpha$, $R$ extends to a holomorphic function of $\eta$ in $\operatorname{Re}\eta>0$.

**Lemma E.1** (bounds on $R$)**.** *Let $s_1=e^{-2\lambda+4\eta}$, $s_2=e^{-\lambda+2\eta}$ (both $<1$ here). Then*

$$R_1\le\frac{16\eta^2 s_1}{(1-s_1)^2(1-e^{-2\lambda})},\qquad R_2\le\frac{4\eta^2 s_2}{(1-s_2)^2(1-e^{-\lambda})},$$

*and for complex $z$ with $x=\operatorname{Re}z>0$ (and $\lambda,s_1,s_2$ evaluated at $x$), $|R(z)|\le\frac{4|z|(1+2|z|)s_1}{(1-s_1)^2(1-e^{-2\lambda})}+\frac{4|z|^2 s_2}{(1-s_2)^2(1-e^{-\lambda})}$. Hence $|\partial_\eta R(\eta)|\le\max_{|z-\eta|=1}|R(z)|$ (Cauchy's estimate) for $\eta>1$.*

*Proof.* Use $\sin 2\alpha\le 2\alpha$, $\sin^2\alpha\le\alpha^2$, $\sinh y\le ye^y$, $1+\cosh y\le 2e^y$ ($y\ge0$), drop the negative term of $R_1$, bound $1/(1-e^{-kn\lambda})\le 1/(1-e^{-k\lambda})$ and sum $\sum nq^n=q/(1-q)^2$. For complex $z$: $|e^{-n\lambda(z)}|=e^{-n\lambda(x)}$, $|\sinh 4nz|\le e^{4nx}$, $|\cosh 2nz|\le\cosh 2nx$, and $4|z|e^{4nx}+8n|z|^2\le 4|z|(1+2|z|)\,ne^{4nx}$. □

**Lemma E.2** (dense rows: region C1)**.** *If $\eta\ge 7$ (i.e. $\ell_u\le\pi\alpha/7$) and $\alpha\le\pi/4$, then $\mathcal{F}(\pi)<0$.*

*Proof.* $\mathcal{F}\le\mathcal{F}^{\mathrm{mod}}$, and the terms $-\frac{\alpha\sin 2\alpha}{\ell_u}$, $-\frac{\cos 2\alpha}{4}$ are $\le 0$. Also $\frac{\pi\sin 2\alpha}{2\ell_u}\le\frac{\pi\sin\alpha}{\ell_u}$ and $\frac{\pi^2\sin^2\alpha}{4\ell_u^2}=\frac{\pi\sin\alpha}{\ell_u}\cdot\frac{\eta}{4}\operatorname{sinc}\alpha\ge\frac{\pi\sin\alpha}{\ell_u}\cdot\frac{\eta}{2\pi}$ (as $\operatorname{sinc}\alpha\ge 2\sqrt{2}/\pi$). With $\coth 2\eta\le\coth\eta\le\coth^2\eta$, $\mathcal{F}^{\mathrm{mod}}\le\frac{\pi\sin\alpha}{\ell_u}\coth^2\eta\,(1-\frac{\eta}{2\pi})<0$ since $7>2\pi$. □

**Lemma E.3** (region C2)**.** *Let $t = 0.25/\pi$ and $h(\eta) = \eta\coth 2\eta - \frac{\eta^2}{4}\coth^2\eta - \frac{1}{4}$. (a) If $\alpha \le t\eta$, then $\mathcal{F}^{\mathrm{mod}} \ge h(\eta) - \eta^2\big[\frac{2t^2}{3}(\frac{1}{2}+\eta) + \frac{2t}{\pi}\big]$. (b) $h(\eta)/\eta^2 \ge \frac{1}{2} - \frac{37}{180}\eta^2$ for $0 < \eta < \pi/2$.*

*Proof.* (a) $\mathcal{F}^{\mathrm{mod}} - h = \eta\coth 2\eta(\operatorname{sinc} 2\alpha - 1) - \frac{\eta^2}{4}\coth^2\eta(\operatorname{sinc}^2\alpha - 1) - \frac{\eta}{\pi}\sin 2\alpha + \frac{1-\cos 2\alpha}{4}$. The second and fourth terms are $\ge 0$; $1 - \operatorname{sinc} x \le x^2/6$ and $\eta\coth 2\eta \le \frac{1}{2} + \eta$ bound the first by $-(\frac{1}{2}+\eta)\frac{2\alpha^2}{3}$, and $\sin 2\alpha \le 2\alpha$ bounds the third by $-\frac{2\eta\alpha}{\pi}$; then use $\alpha \le t\eta$. (b) With $A(x) = x\coth x$, $h/\eta^2 = [2A(2\eta) - A(\eta)^2 - 1]/(4\eta^2)$, and for $0 < x < \pi$ the alternating series $A(x) = 1 + \frac{x^2}{3} - \frac{x^4}{45} + \frac{2x^6}{945} - \dots$ has decreasing terms, so $A(2\eta) \ge 1 + \frac{4\eta^2}{3} - \frac{16\eta^4}{45}$ and $A(\eta) \le 1 + \frac{\eta^2}{3}$; hence $2A(2\eta) - A(\eta)^2 - 1 \ge 2\eta^2 - \frac{37}{45}\eta^4$. □

**The certificate for $\ell_u < 0.25$** (`code/region_C_cert.py`, Arb ball arithmetic, about one second). Every zero with $\ell_u < 0.25$ has $\alpha < \pi/4$ and one of: *C1* $\eta \ge 7$: impossible by Lemma E.2. *C2* $\eta \le 2.5$: then $\alpha = \eta\ell_u/\pi \le t\eta$; on $(0, 0.05]$ Lemma E.3(b), and on $[0.05, 2.5]$ 117 mean-value enclosures of $h/\eta^2$, give $h/\eta^2 - [\frac{2t^2}{3}(\frac{1}{2}+\eta) + \frac{2t}{\pi}] \ge 0.0222$, while Lemma E.1 gives $R/\eta^2 \le 3.1\cdot 10^{-32}$ (here $\lambda \ge 2\pi^2/0.25$); so $\mathcal{F} > 0$, no zero. *C3* $2.5 \le \eta \le 7$: then $\alpha \le 7\cdot 0.25/\pi < 0.557$, and Lemma E.1 gives $R \le 1.7\cdot 10^{-8}$ and $|\partial_\eta R| \le 2.4\cdot 10^{-4}$ on the whole region; 113 boxes in $(\alpha, \eta)$ each satisfy $\mathcal{F}^{\mathrm{mod}} - R_{\max} > 0$ (no zero), or $\mathcal{F}^{\mathrm{mod}} < 0$ (no zero), or $\partial_\eta\mathcal{F}^{\mathrm{mod}} + |\partial_\eta R|_{\max} < 0$ (any zero is transversal).

**Lemma E.4** (dense rows, any curvature; **[Lean]**)**.** *For fixed $\alpha \in (0, \pi)$, $\ell_u^2\,\mathcal{F}(\pi) \to -\frac{\pi^2\sin^2\alpha}{4}$ as $\ell_u \to 0^+$. Hence $\mathcal{F}(\pi) < 0$, and the row is unstable, for all sufficiently small strip spacings. Lean:* `HyperbolicRings.Rows.tendsto_F_small, F_neg_small`.

*Proof.* By Lemma 6.2 the $j$-th term of $\ell_u^2\mathcal{F}(\pi)$ is $\ell_u^2\sigma\mathfrak{g}_{\mathrm{o}}(c_j)$ ($j$ odd) or $\ell_u^2\sigma\mathfrak{g}_{\mathrm{e}}(c_j)$ ($j$ even). As $\ell_u \to 0$ these tend to $-2\sigma/j^2$ and $0$ ($(c_j - 1)/\ell_u^2 \to j^2/2$, $c_j \to 1$), and both are bounded by $12/j^2$, using $c_j - 1 \ge j^2\ell_u^2/2$ and $D \ge \max(c_j - 1, 2\sigma)$. By dominated convergence the limit is $-2\sigma\sum_{j\ \mathrm{odd}} j^{-2} = -\pi^2\sigma/4$. □

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